Basics of dynamic behaviour of metal materials

Online preparation

Assoc. Prof. Vegard Aune

April 7, 2026

Course on:
Deep Learning–Driven Design under Dynamic Loads

- **Online part** (common to all participants): April 7, 9, 14, 16 - from 2:30 PM to 4:00 PM
- Core course at Polimi: April 20-24 - from 9:00 AM to 6:00 PM
- Core course at NTNU: June 23-24 - from 9:00 AM to 4:00 PM

Students attend on-site only when lessons are held at their home university, joining remotely for all other sessions.

ENHANCE

POLITECNICO MILANO 1863

NTNU
Norwegian University of Science and Technology

Online part

The course will begin with a series of **online preparatory lectures**, which introduce key concepts required for the in-person sessions:

- **Session 1** – Basics of dynamic behavior of metal materials (Tuesday, April 7)
- **Session 2** – Basics of FE simulations (Thursday, April 9)
- **Session 3** – Basics of blast loading (Tuesday, April 14)
- **Session 4** – Basics of Deep Learning (Thursday, April 16)

These sessions **provide the foundations** for the hands-on activities during the course:

- Application of deep learning to both fluid-structure interaction surrogate modeling and lattice structures design

All **course materials** (including links to the online sessions, slides, notes, and coding notebooks) will be available in the **GitHub repository**:

- https://github.com/lucalomazziPolimi/Pia-Enhance-Course.git

Basics of dynamic behaviour of (metal) materials

Online preparation

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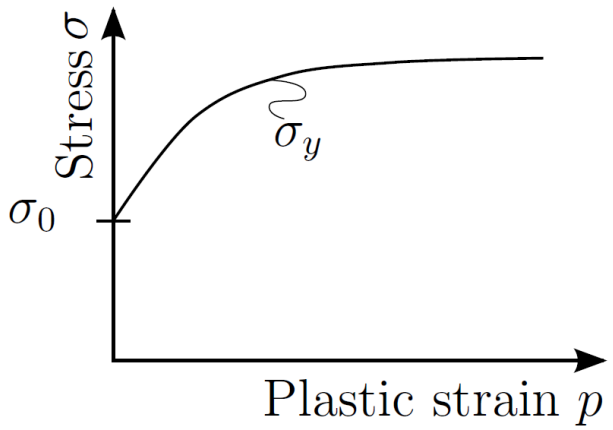
Metals vs. polymers

Understanding the **fundamental difference** between how metals and polymers behave dynamically is essential for selecting materials in impact and blast applications.

- The governing physics are very different, and so are the modeling requirements.

Property / Feature	Metals	Polymers
Deformation mechanism	Dislocation motion in crystal lattice	Molecular chain motion / disentanglement
Strain-rate sensitivity	Moderate (log-linear)	Very strong (can change character)
Temperature sensitivity	Moderate	Very strong
Wave speed	~5,000 m/s	~1,000–2,000 m/s
Constitutive framework	Elastic-viscoplastic	Hyperviscoelastic / viscoplastic
Failure mode at high rate	Adiabatic shear bands, spalling	Brittle fracture, debonding

Modelling of metals



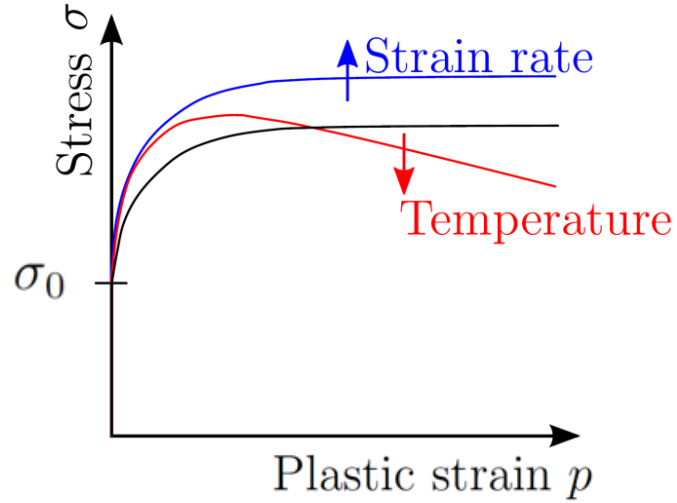
- Stresses updated based on the yield function

$$f(\boldsymbol{\sigma}) = \sigma_{eq} - \sigma_y$$

- Isotropic and pressure-insensitive yield criterion

$$\sigma_{eq}(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2}\boldsymbol{\sigma}' : \boldsymbol{\sigma}'}$$

- Modified Johnson-Cook (stress versus strain)



$$\sigma_y = \left[\sigma_0 + \sum_{i=1}^{2} Q_i(1-\exp(-C_i p))\right]\left[1+\frac{\dot{p}}{\dot{p}_0}\right]^c \left[1-T^{*m}\right]$$

Johnson and Cook (1983)
Aune et al. (2017)

Modeling dynamic behaviour of polymers

Constitutive modeling of polymers under dynamic loading is considerably **more complex than for metals**.

- The combination of large deformation, viscoelasticity, viscoplasticity, and strong thermal coupling requires specialised rheological models.

Required model capabilities

- Large strain kinematics (finite deformation)
- Strain-rate dependence across many decades
- Temperature coupling (adiabatic self-heating)
- Viscoelastic and viscoplastic contributions
- Failure and fracture criteria

Common modeling approaches

- **Hyperelastic models** (Neo-Hookean, Mooney-Rivlin, Arruda-Boyce) - for rubbers at large strains
- **Hyperviscoelastic models** (Ogden + Prony series, Bergström-Boyce) - rate-dependent large deformation
- **Viscoplastic models** (Arruda-Boyce + flow rule) - for glassy and semicrystalline polymers
- **Coupled thermo-mechanical models** - for high-rate adiabatic applications

Contents

- Elasticity
- Viscoelasticity
- Plasticity
- Viscoplasticity
- Large deformations
- Metal foams

Elasticity

Characteristics

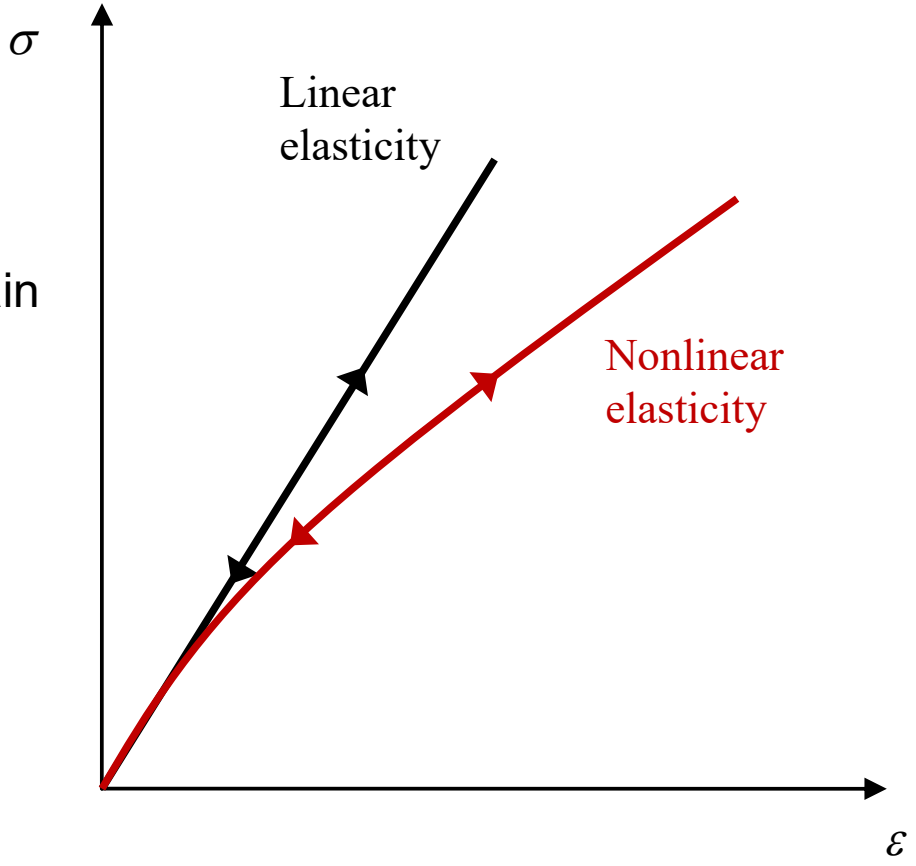
The material returns to its original shape after the stress that induced the strain is removed

Arch bridges in Gundián, Galicia (Spain). 1956 and 2008.

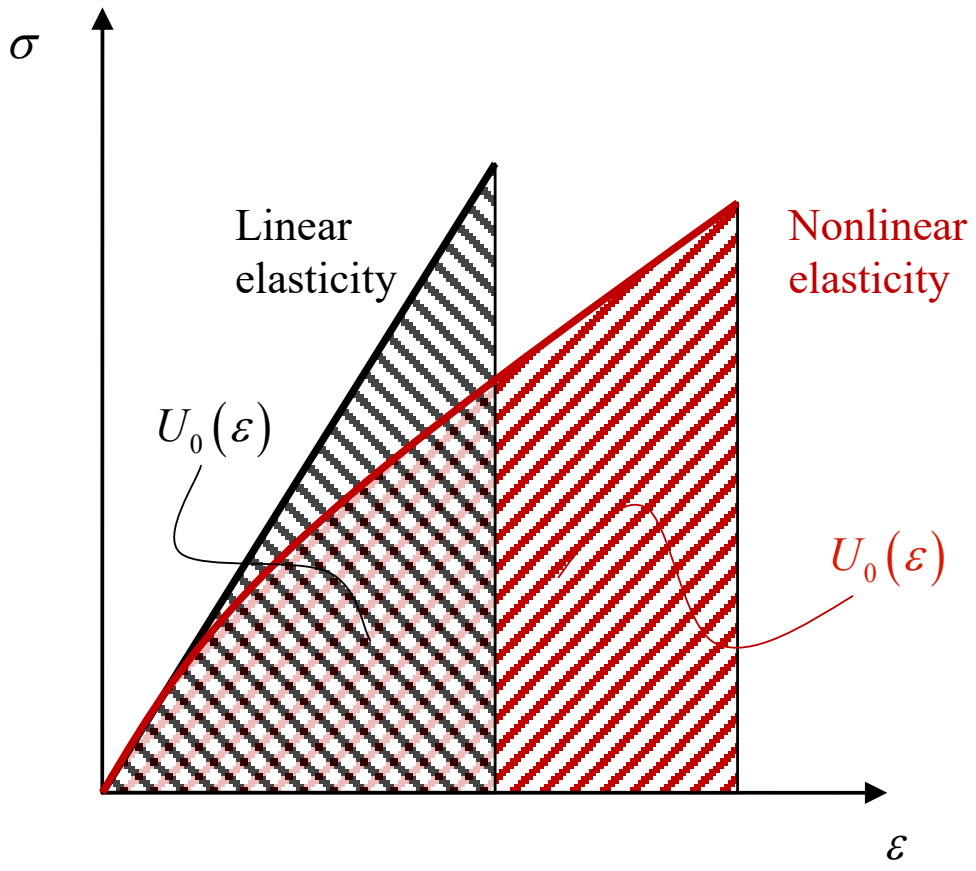
Characteristics

The material returns to its original shape after the stress that induced the strain is removed

- Reversible deformations
  - The body returns to original shape after unloading
- Path independence
  - A one-to-one correspondence between stress and strain
- No dissipation of mechanical energy
  - The energy is stored in the body as strain energy
- Rate independence
  - The response is independent on the rate of loading



Stored energy



Elastic energy stored in the material per unit volume:

$$U_0(\varepsilon) = \int_0^{\varepsilon} \sigma(\varepsilon) d\varepsilon$$

where $U_0(\varepsilon)$ is the **strain energy function**.

U_0 is a potential:

$$\sigma(\varepsilon) = \frac{\partial U_0(\varepsilon)}{\partial \varepsilon}$$

Small strains: linear elasticity in 1D

- Infinitesimal strains ➔ linear stress-strain relation.

- Linear elastic material (*Hooke's law*)

$$\sigma = E\varepsilon \quad \text{and} \quad \varepsilon = \sigma/E$$

where E is Young's modulus.

- Strain energy function:

$$U_0(\varepsilon) = \int_0^{\varepsilon} \sigma(\varepsilon) d\varepsilon = \int_0^{\varepsilon} E\varepsilon d\varepsilon = \frac{1}{2} E\varepsilon^2$$

- Stored elastic energy of the body:

$$U = \int_V U_0 dV$$

where V is the volume of the solid body.

High service loads, but infinitesimal strains.
~300-500 tonnes

Elastic materials: summary

Linear elastic

- Stress-strain relation

$$\sigma = E\varepsilon \quad \Leftrightarrow \quad \varepsilon = \sigma / E$$

- Strain energy function

$$U_0(\varepsilon) = \tfrac{1}{2} E\varepsilon^2$$

- Stress:

$$\sigma(\varepsilon) = \frac{dU_0}{d\varepsilon} = E\varepsilon$$

Non-linear elastic

- Stress-strain relation:

$$\sigma = \sigma(\varepsilon) \quad \Leftrightarrow \quad \varepsilon = \varepsilon(\sigma)$$

- Strain energy function:

$$U_0(\varepsilon) = \int_0^{\varepsilon} \sigma(\varepsilon)\, d\varepsilon$$

- Stress:

$$\sigma(\varepsilon) = \frac{dU_0}{d\varepsilon}$$

Viscoelasticity

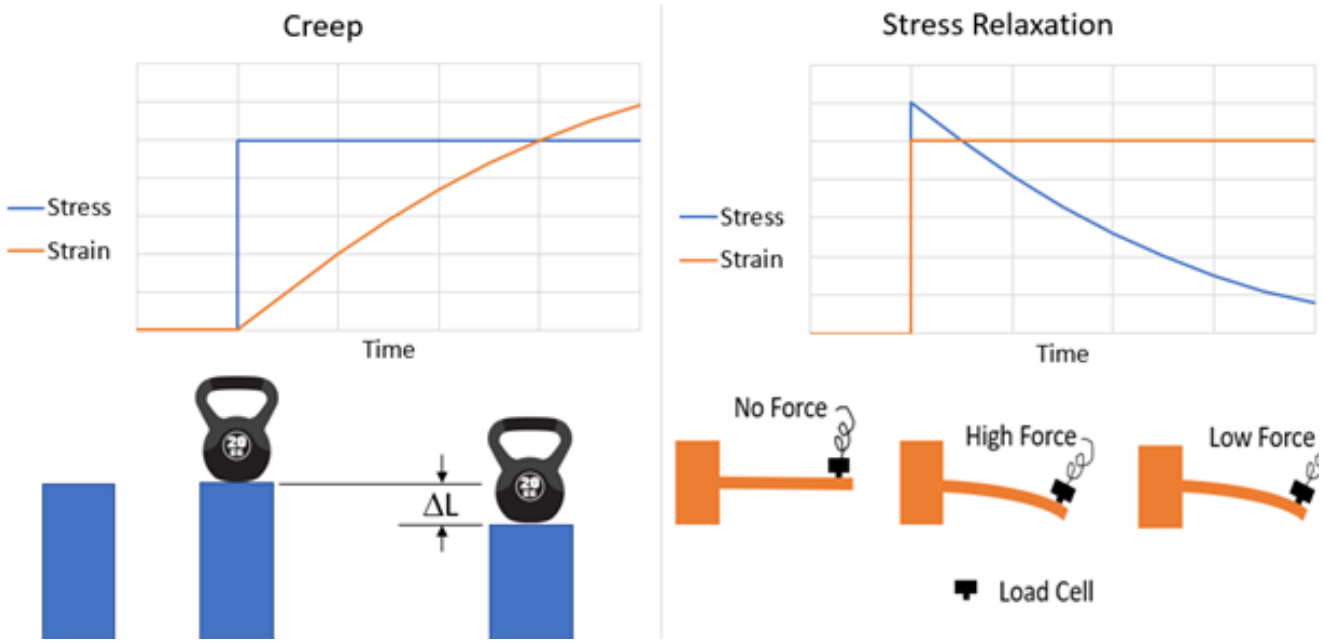
Characteristics

The material exhibits both viscous (fluid-like) and elastic (solid-like) characteristics.

- Reversible/irreversible deformation
- Path dependence
- Energy dissipation
- Rate dependence

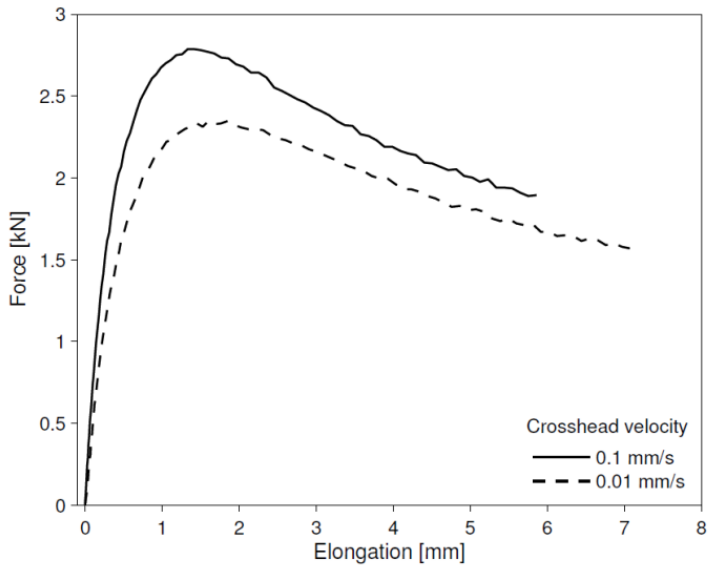
Some phenomena

- Creep:
  - If the stress is held constant, the strain increases with time
- Stress relaxation:
  - If the strain is held constant, the stress decreases with time

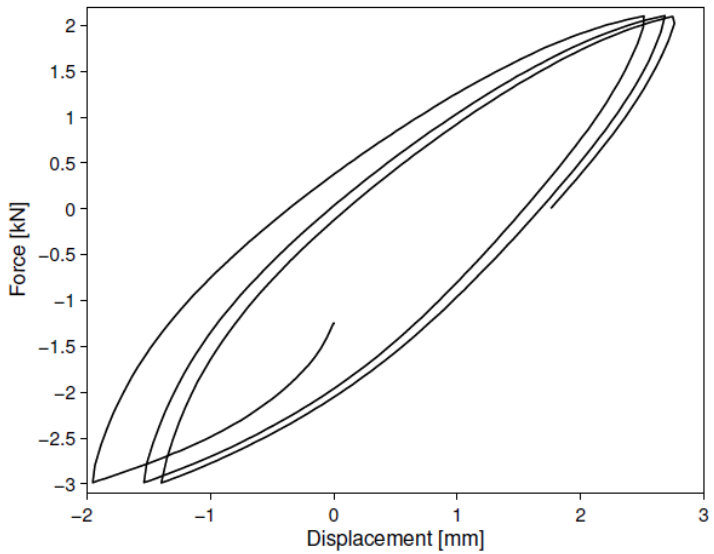


Some phenomena

- Rate-dependent stiffness:
  - The effective stiffness depends on the rate of application of the load

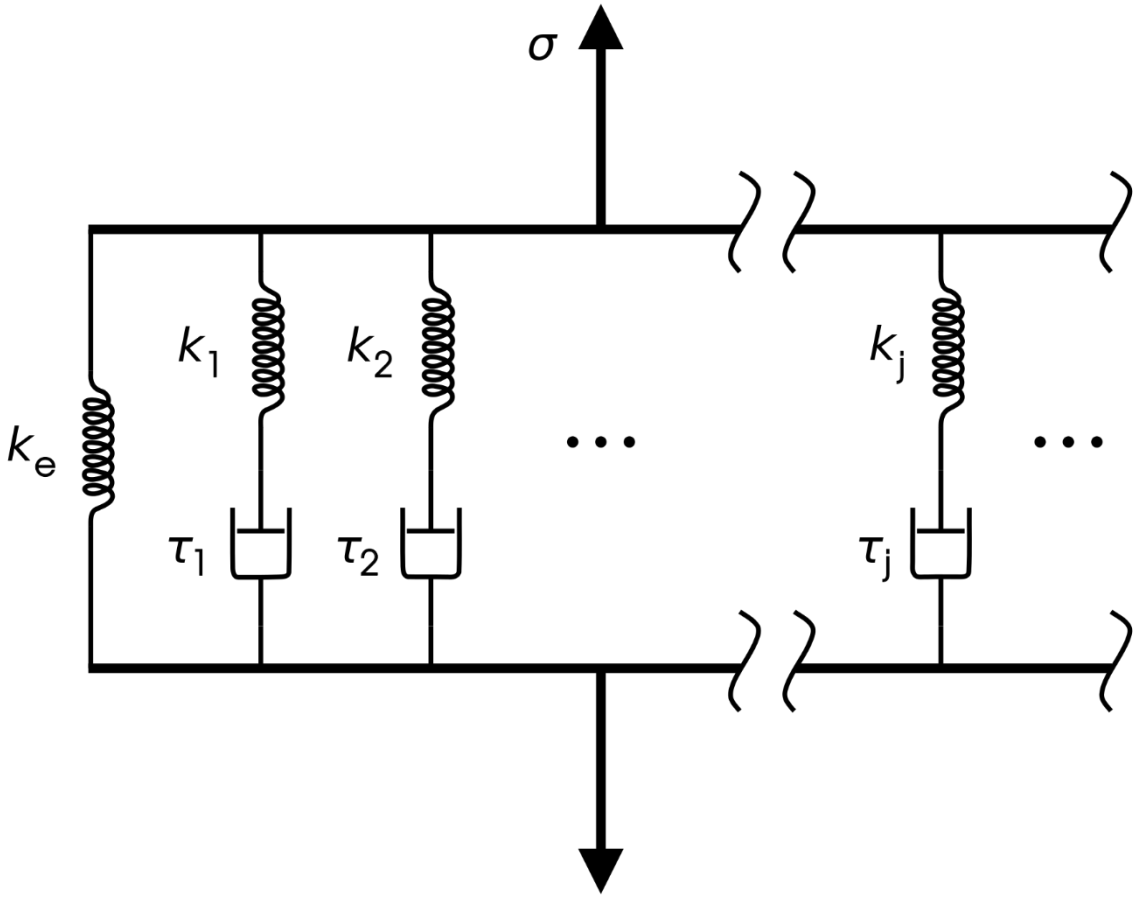


- Hysteresis (dissipation):
  - If cyclic loading is applied, hysteresis occurs, leading to a dissipation of mechanical energy

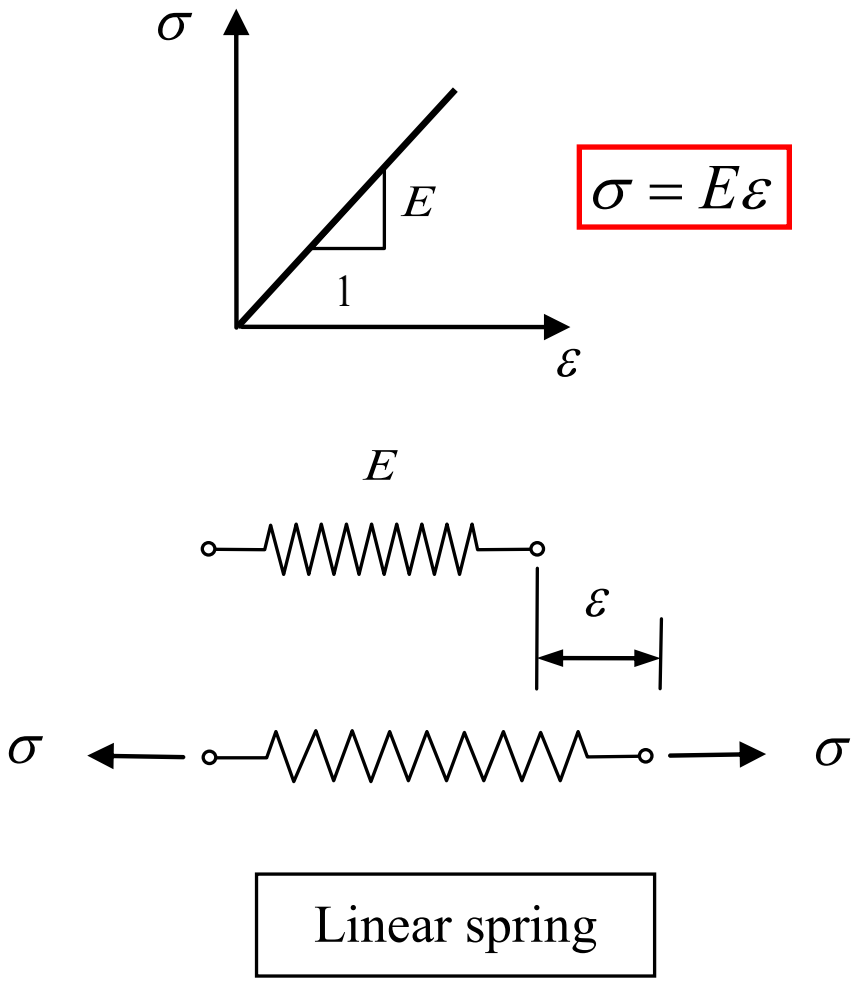


Rheological models

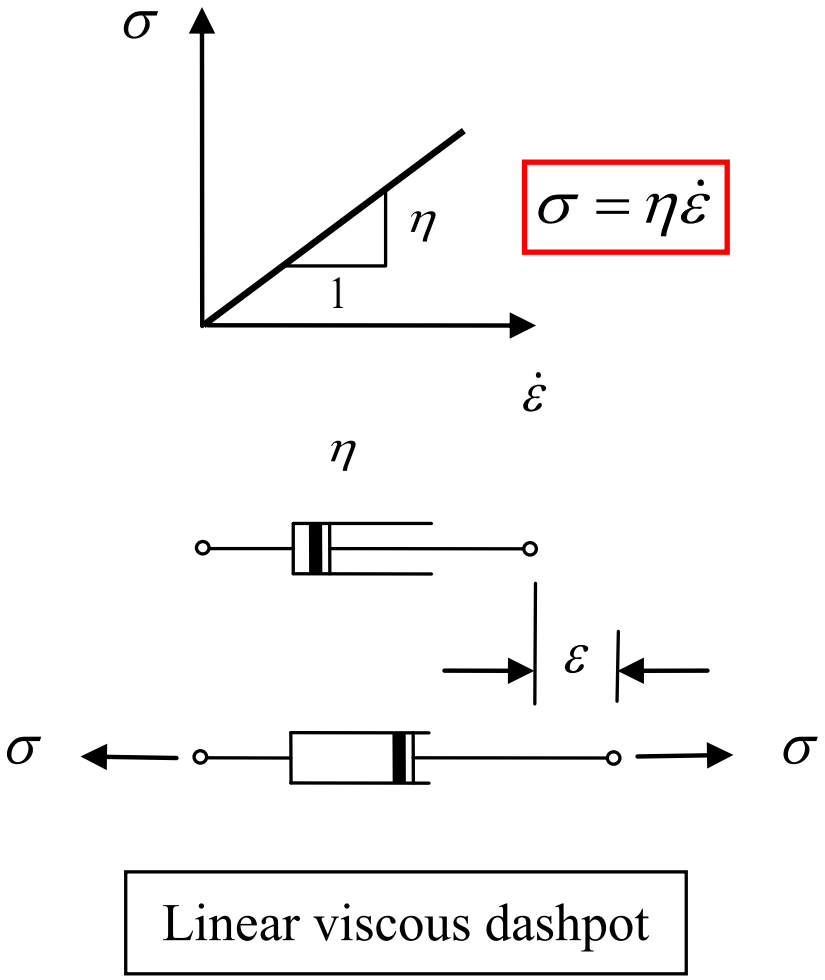
- Viscoelastic materials are usually described with **rheological models** consisting of a combination of **linear springs** and **viscous dashpots**:



Spring and dashpot

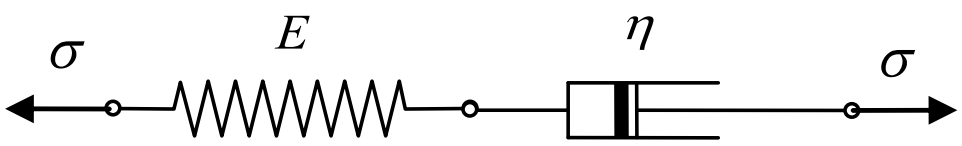


Linear spring



Linear viscous dashpot

Maxwell model

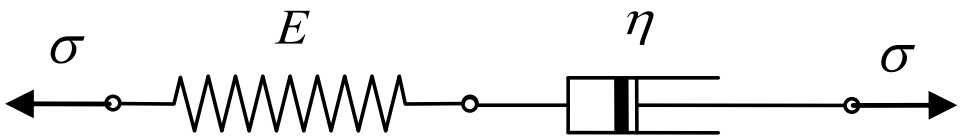


- Strain decomposition (or compatibility): $\varepsilon = \varepsilon^e + \varepsilon^i \Rightarrow \dot{\varepsilon} = \dot{\varepsilon}^e + \dot{\varepsilon}^i$

- Equilibrium: $\sigma = E\varepsilon^e = \eta\dot{\varepsilon}^i \quad \Rightarrow \quad \varepsilon^e = \dfrac{\sigma}{E}, \quad \dot{\varepsilon}^i = \dfrac{\sigma}{\eta}$

- Combining both: differential equation: $\boxed{\dfrac{\dot{\sigma}}{E} + \dfrac{\sigma}{\eta} = \dot{\varepsilon}}$

Maxwell model (2)



$$\boxed{\frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta} = \dot{\varepsilon}}$$

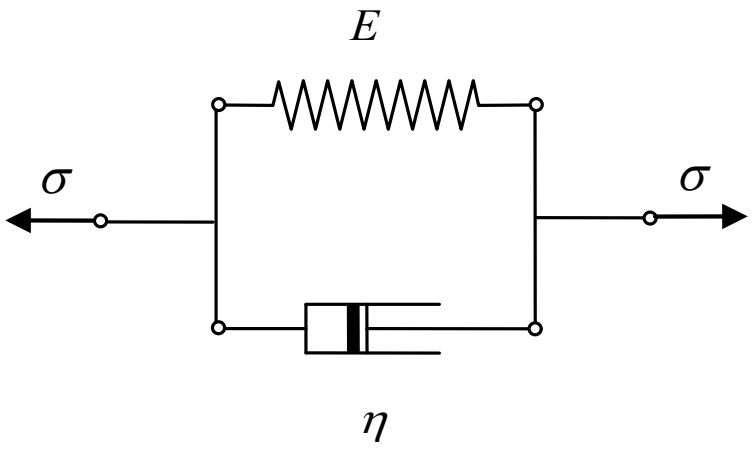
- Stress power per unit volume: $\omega_d = \sigma\dot{\varepsilon} = \sigma\dot{\varepsilon}^e + \sigma\dot{\varepsilon}^i$

- **Inelastic dissipation** (in the dashpot): $\mathfrak{D}_i = \sigma\dot{\varepsilon}^i = \eta\dot{\varepsilon}^i\dot{\varepsilon}^i \geq 0$ Always non-negative!

The elastic power $\sigma\dot{\varepsilon}^e$ is stored in the spring as strain energy, while the viscous power $\sigma\dot{\varepsilon}^i$ related to the viscous dashpot is dissipated and converted into heat.

If $t \to \infty$, then σ goes to 0. Not very realistic for a solid, could work for a fluid.

Kelvin-Voigt model (1)

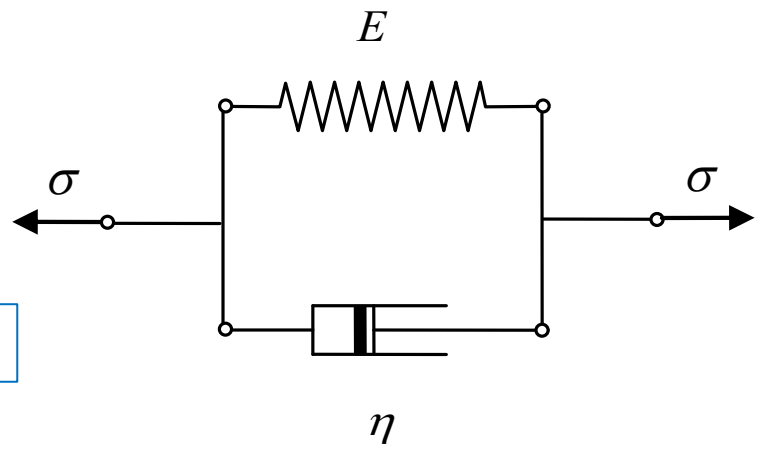


- Strain decomposition (or compatibility): both branches move in parallel.

- Equilibrium: $\sigma = \sigma^e + \sigma^i$

- Differential equation: $\sigma^e = E\varepsilon, \quad \sigma^i = \eta\dot{\varepsilon} \quad \Rightarrow \boxed{\sigma = E\varepsilon + \eta\dot{\varepsilon}}$

Kelvin-Voigt model (2)

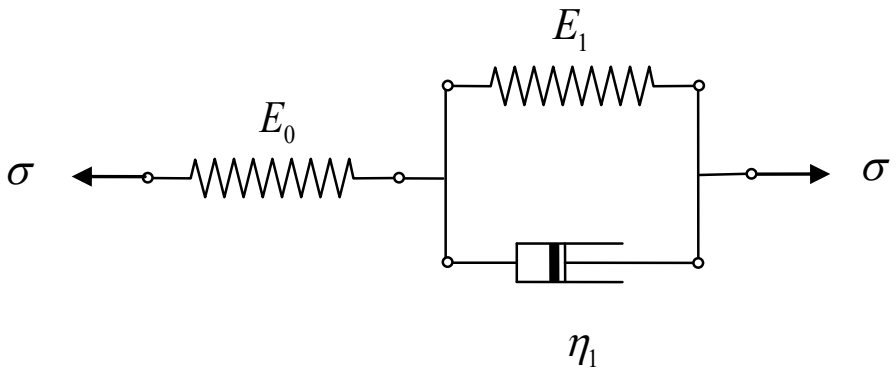


$$\boxed{\sigma = E\varepsilon + \eta\dot{\varepsilon}}$$

- Stress relaxation: We know $\varepsilon(t) = \varepsilon_0 =$ constant ($\dot{\varepsilon} = 0$), suddenly applied.
- BUT it is not possible to apply a sudden strain to the model!
- Even if we started from a certain strain, the stress would be constant

The Kelvin-Voigt model is unable to describe stress relaxation

Standard linear model



Internal variable used to describe inelastic (rate dependent) material behaviour

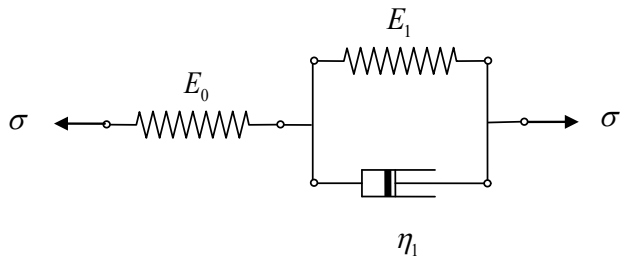
- Strain decomposition: $\varepsilon = \varepsilon^e + \varepsilon^i$
- Equilibrium: $\sigma = E_0 \varepsilon^e \quad \wedge \quad \sigma = E_1 \varepsilon^i + \eta_1 \dot{\varepsilon}^i$
- Constitutive relations: $\boxed{\varepsilon = \dfrac{\sigma}{E_0} + \varepsilon^i, \quad \dot{\varepsilon}^i = \dfrac{1}{\eta_1}\left(\sigma - E_1 \varepsilon^i\right)}$

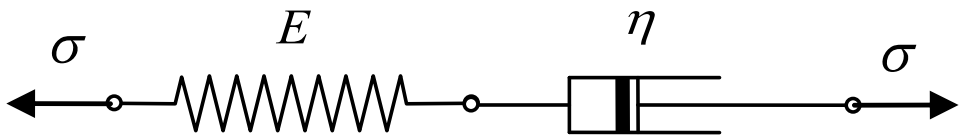
We could combine both equations, but it is a good idea to leave the $\dot{\varepsilon}^i$ part uncoupled

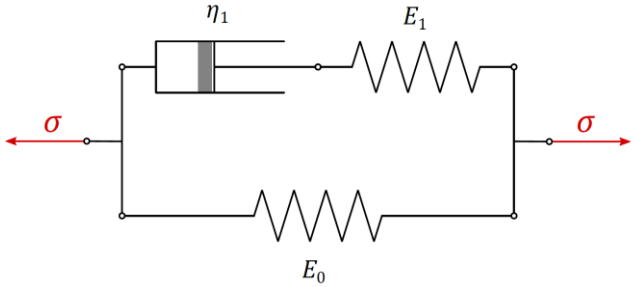
The standard linear model is able to describe stress relaxation

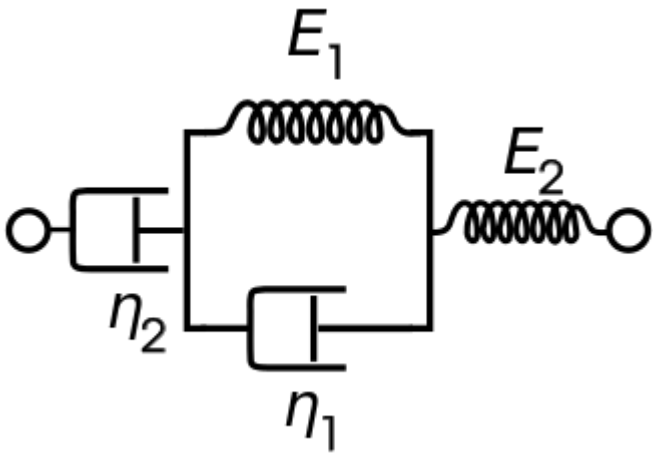
How to identify solid and fluid rheological models?

"If you can walk on springs, it is a solid"









Plasticity

Physical reasons

- We limit ourself to **phenomenological modelling**. This means that the characteristics of the materials are modelled based on **experimental observations** without trying to model the underlying physical mechanisms (i.e. the microstructure).

- For metals, the atomic structure changes size and shape when forces are applied to the system during **elastic deformation**. When these forces are removed, the structure goes back to the original state.

- **Plastic deformation** is mainly caused by plastic slip of the crystal lattice caused by **movement of dislocations** (or defects), while **work hardening** is caused by an **increase of the dislocation density** that hinders the plastic flow.

- Since **plastic slip is a shear-driven process**, we normally assume that plasticity in metals are pressure insensitive.

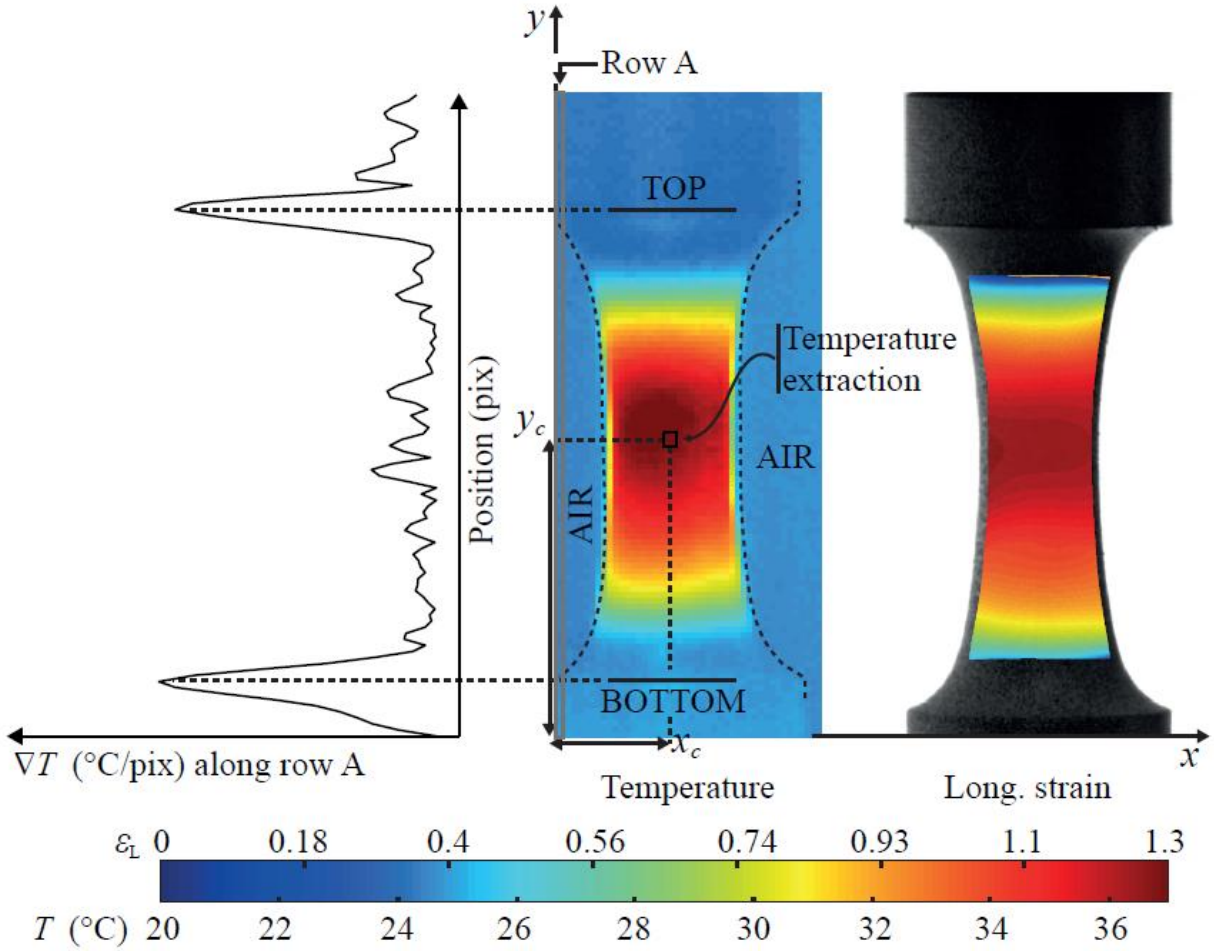
- Note that the theory of plasticity is mainly based on metal plasticity.

Characteristics

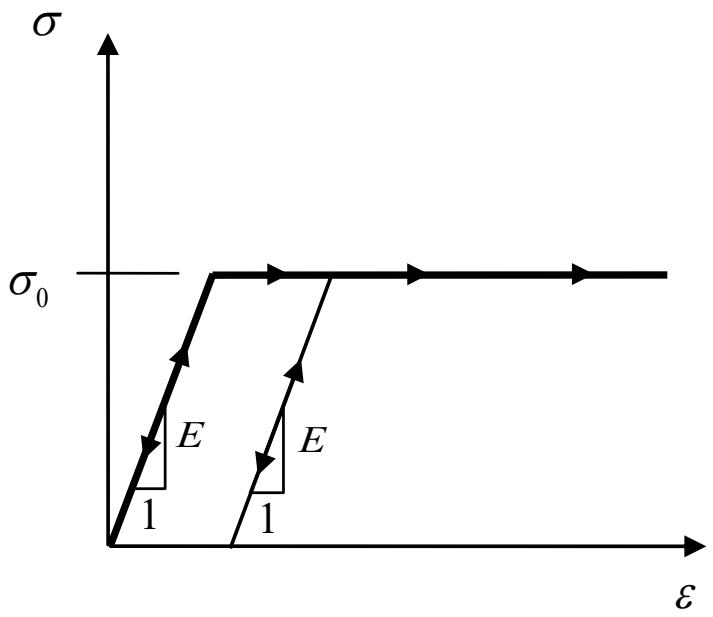
Plasticity describes the deformation of a material experiencing irreversible changes of shape in response to applied forces.

- Irreversible deformation
- Path dependence
- Rate independence
- Energy dissipation

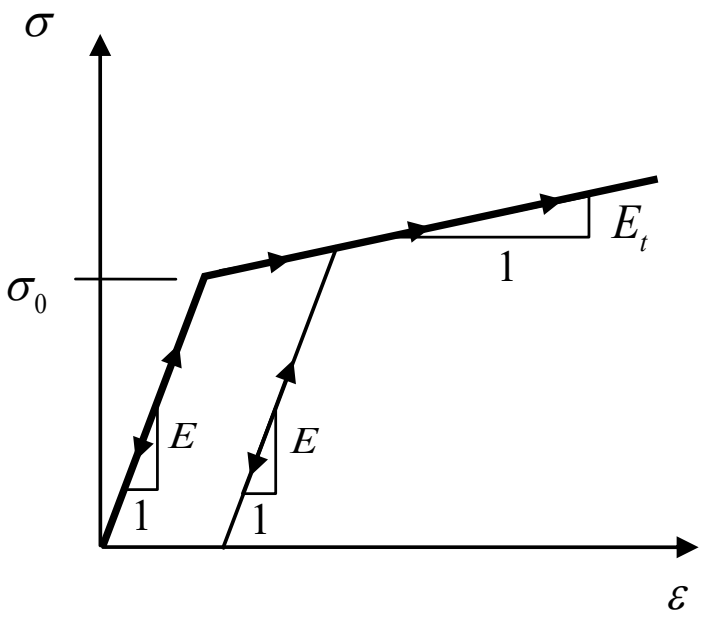
Energy dissipation: heat



Idealized behaviour loading/unloading



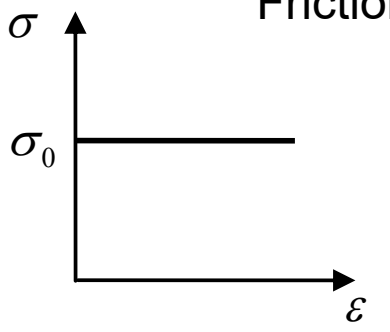
Elastic-perfectly plastic

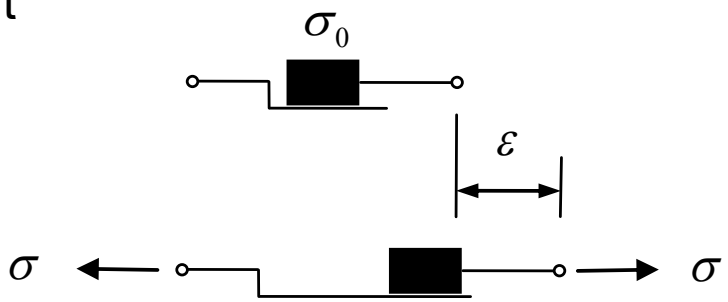


Elastic-linear work-hardening

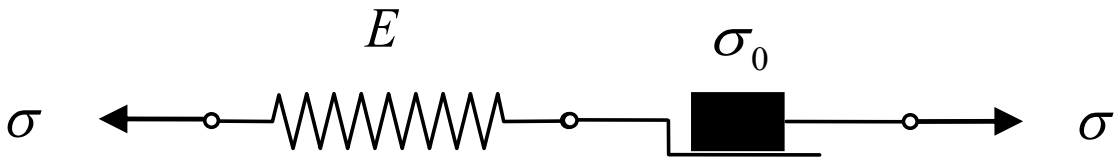
Rheological model

Friction element

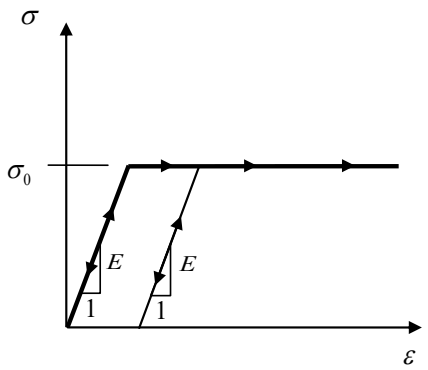




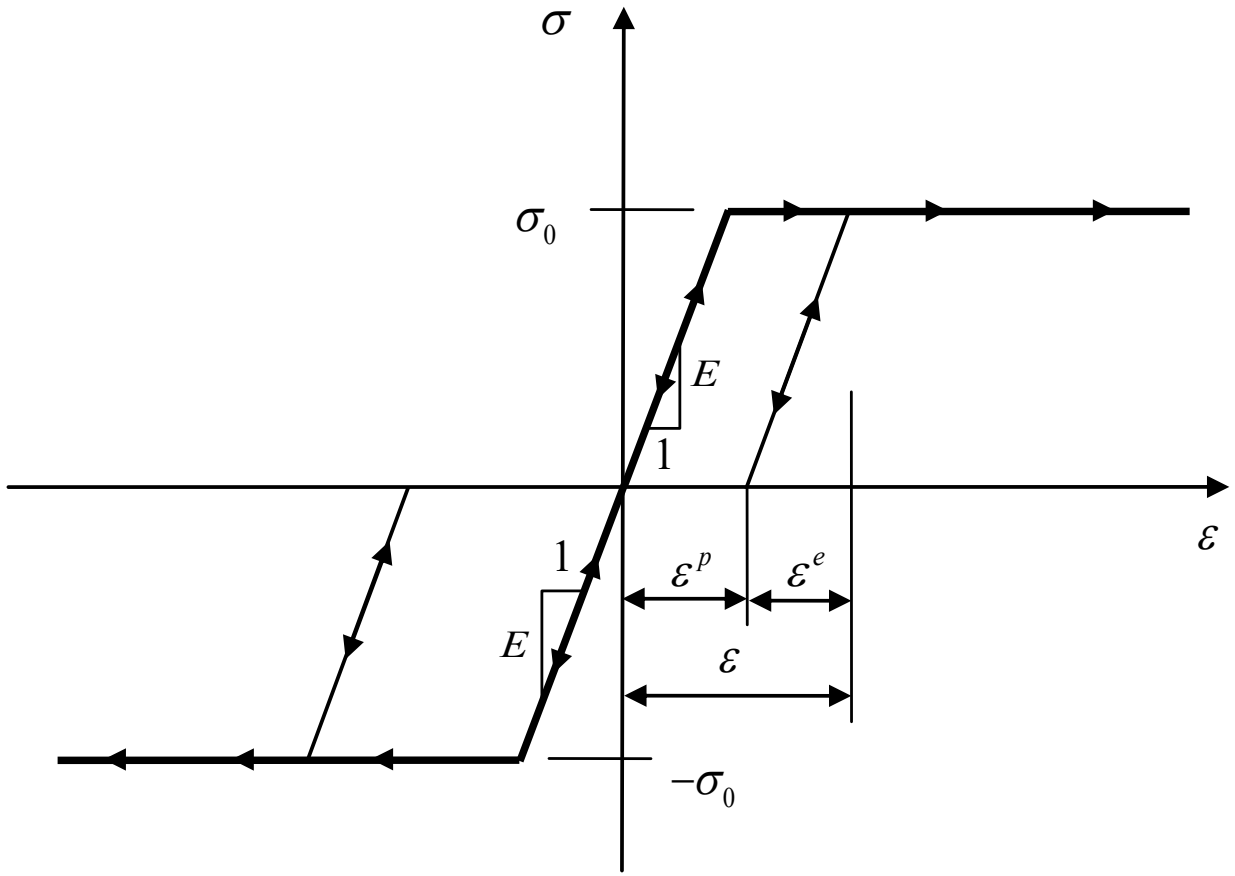
Rheological model for elastic-perfectly plastic material



$$\varepsilon = \varepsilon^e + \varepsilon^p = \frac{\sigma}{E} + \varepsilon^p$$



Yield condition for elastic-perfectly plastic material



$$\boxed{f = |\sigma| - \sigma_0 = 0}$$

Interpretation:

$f < 0$ → elastic domain

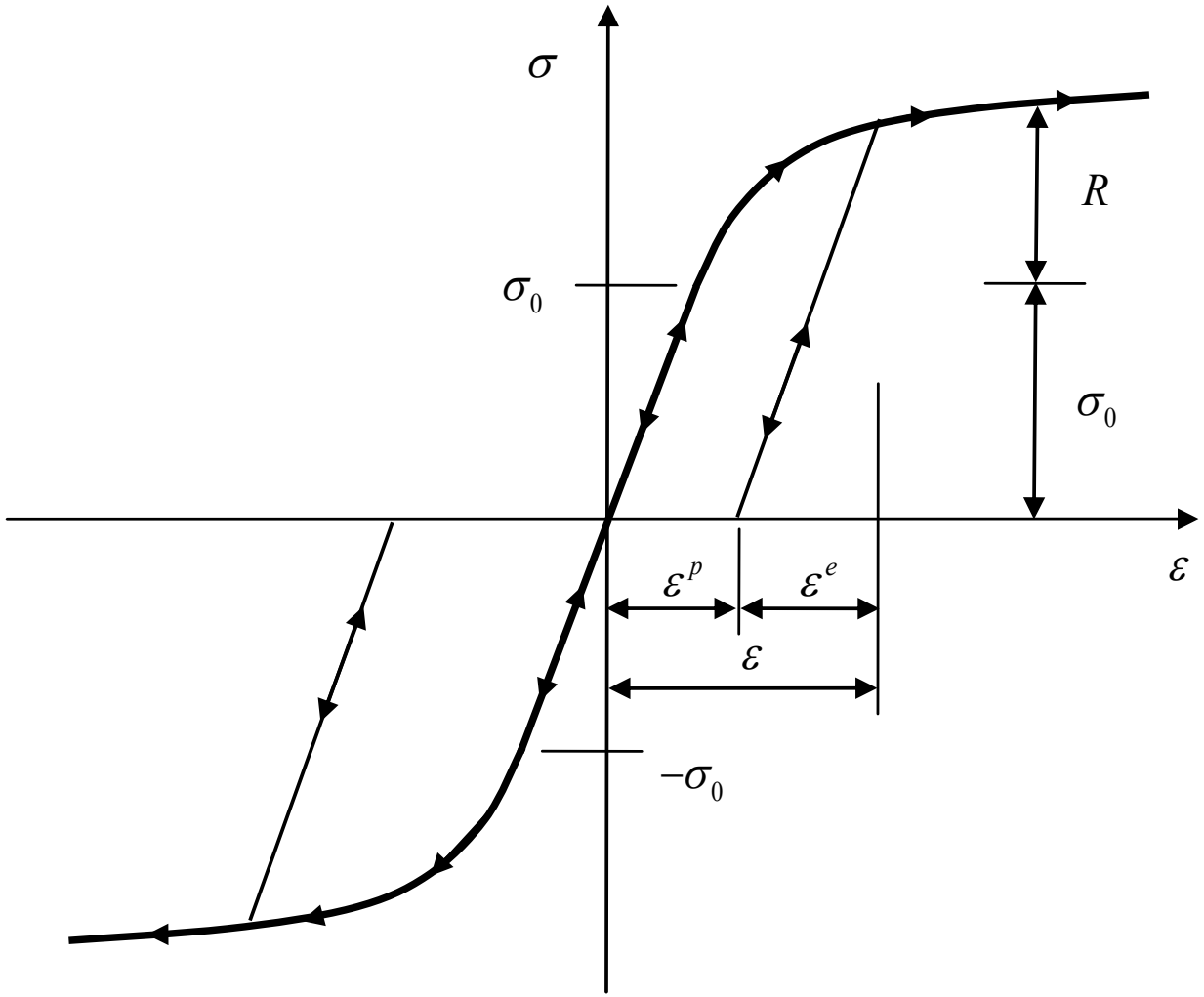
$f = 0$ → plastic domain

$f > 0$ → inadmissible (not allowed)

Equivalent stress = magnitude of stress:

$$\sigma_{\mathrm{eq}} = |\sigma| = \sqrt{\sigma^2}$$

Isotropic hardening: yield function



Yield function

$$f(\sigma, R) = |\sigma| - (\sigma_0 + R) = 0$$

Hardening rule

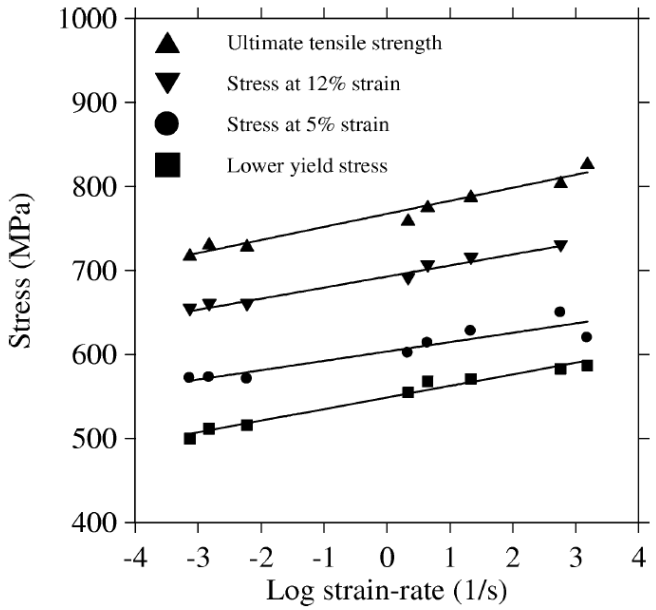
$$R = R(p) \quad \text{or} \quad \dot{R} = H_R \dot{p}$$

where H_R is the isotropic hardening modulus.

Viscoplasticity

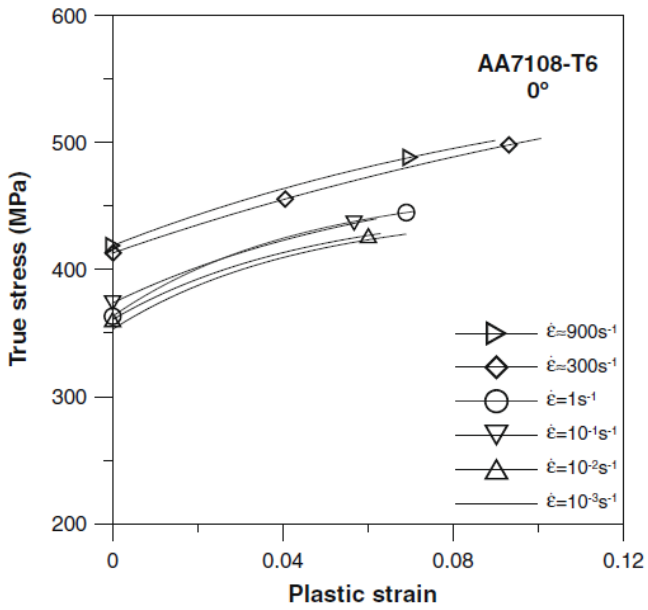
Motivation

Weldox 460 E steel



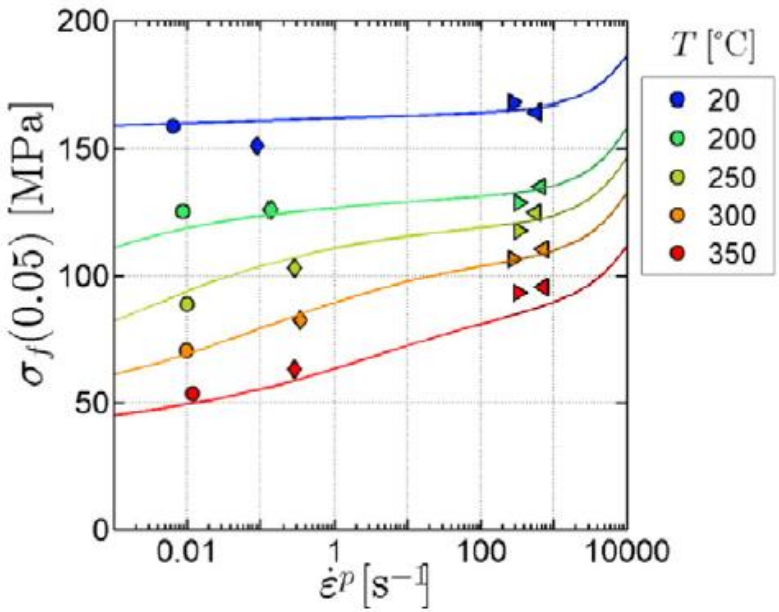
Børvik et al. (2001), EJM-A/Solids.

Aluminium alloy AA7108



Chen et al. (2009), EJM-A/Solids.

Aluminium alloy AA6082

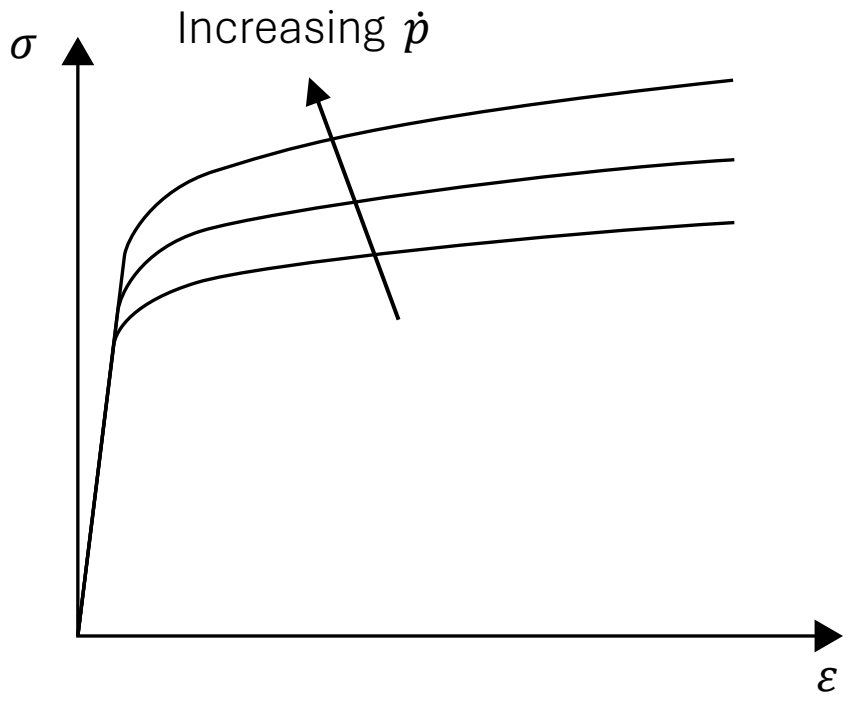


Vilamosa et al. (2016), Mat. & Des.

Characteristics

Viscoplasticity describes the rate-dependent inelastic behaviour of a material

- Irreversible deformation
- Path dependence
- Energy dissipation
- **Rate dependence**

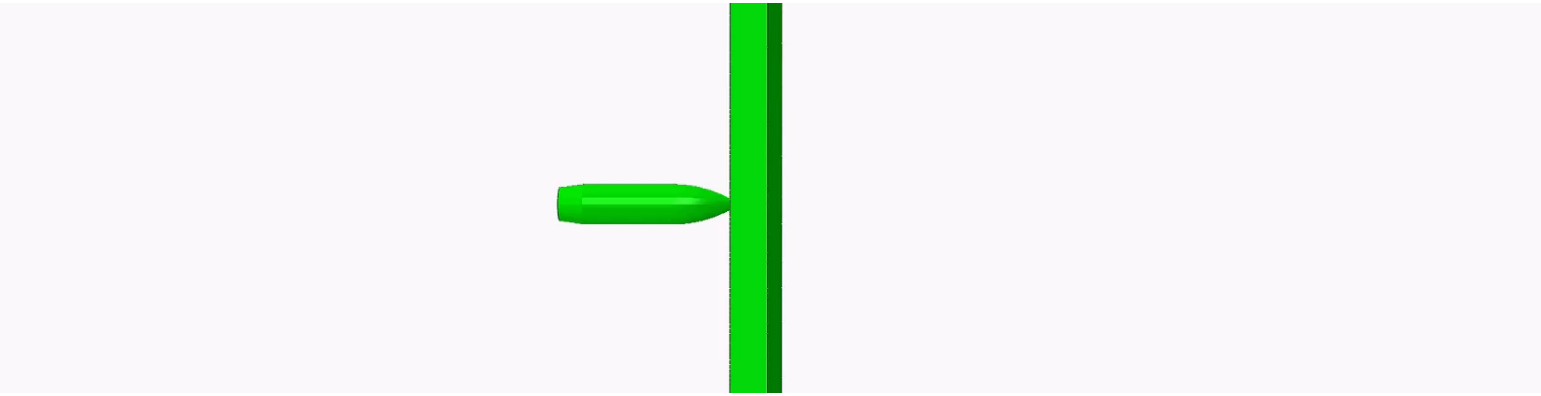


Important applications

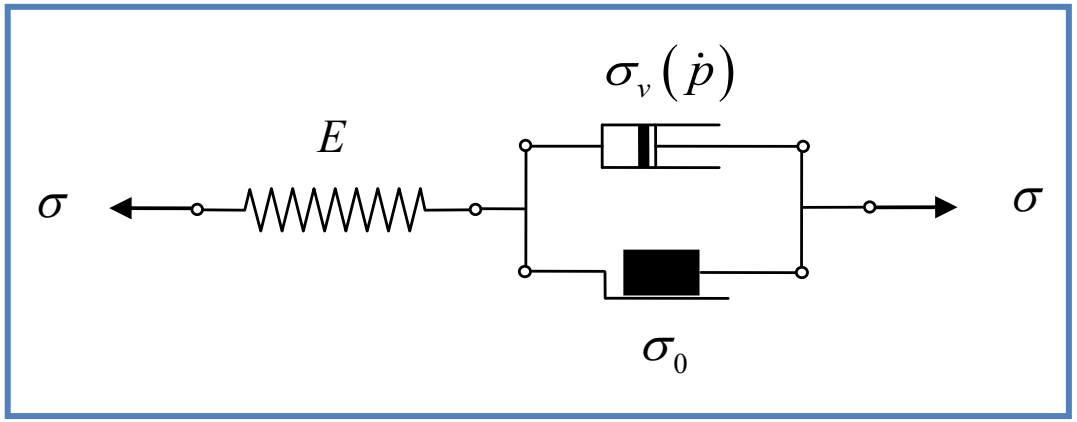
- Creep
- Stress relaxation
- Forming operations
- Structural impact problems

AA7108-T6

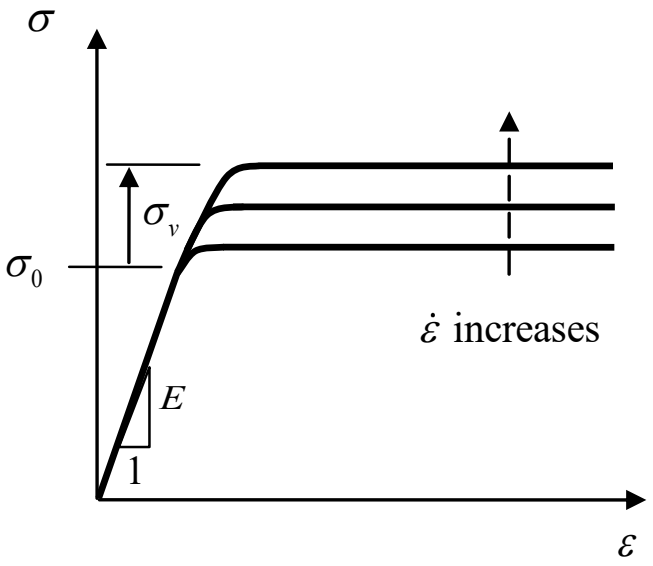
Quasi-static Impact



Rheological model



Elastic – perfectly viscoplastic material



- We add a viscous stress to account for strain rate sensitivity:

$$|\sigma| = \sigma_0 + \sigma_v(\dot{p})$$

- Strain decomposition:

$$\varepsilon = \varepsilon^e + \varepsilon^p$$

- Hooke's law:

$$\varepsilon^e = \frac{\sigma}{E}$$

Viscoplasticity with isotropic hardening

- Strain decomposition:

$$\varepsilon = \varepsilon^e + \varepsilon^p$$

- Hooke's law:

$$\varepsilon^e = \sigma/E \quad \Leftrightarrow \quad \sigma = E\varepsilon^e$$

- Yield function:

$$f = |\sigma| - (\sigma_0 + R(p))$$

- Work hardening:

$$R = R(p) \quad \text{or} \quad \dot{R} = h_R \dot{p}$$

- Equivalent stress:

$$\sigma_{eq} = |\sigma| = \sigma_0 + R(p) + \sigma_v(\dot{p})$$

- Stress power:

$$\omega_d = \sigma\dot{\varepsilon} = \sigma\dot{\varepsilon}^e + \sigma\dot{\varepsilon}^p$$

- Plastic dissipation:

$$\mathfrak{D}_i = \sigma\dot{\varepsilon}^p \geq 0$$

- Flow rule:

$$\dot{\varepsilon}^p = \mathrm{sgn}(\sigma)\,\dot{\lambda} = \dot{\lambda}\frac{\partial f}{\partial \sigma}, \quad \dot{\lambda} \geq 0$$

- Equivalent plastic strain:

$$\sigma\dot{\varepsilon}^p = \sigma_{eq}\dot{p} \quad \Rightarrow \quad \dot{p} = \dot{\lambda}$$

$$p = \int \dot{\lambda} dt = \int \dot{p} dt$$

Important differences compared to rate-independent plasticity

- For plastic deformation to take place, the equivalent stress must be larger than the yield stress (outside the elastic domain):

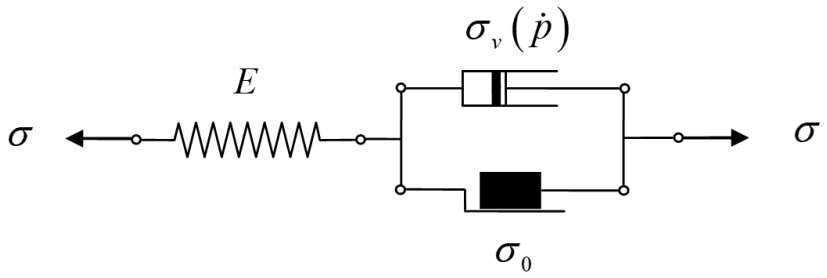
$$f = |\sigma| - \left(\sigma_0 + R(p)\right) = \sigma_v(\dot{p}) > 0 \;\Rightarrow\; \text{plastic flow}$$

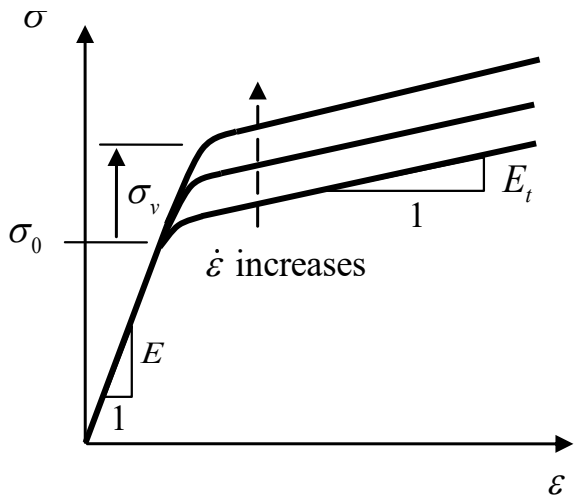
This means that, in terms of the yield function, we have:

$$f \leq 0 \quad \Rightarrow \quad \text{elastic response}$$
$$f > 0 \quad \Rightarrow \quad \text{plastic response}$$

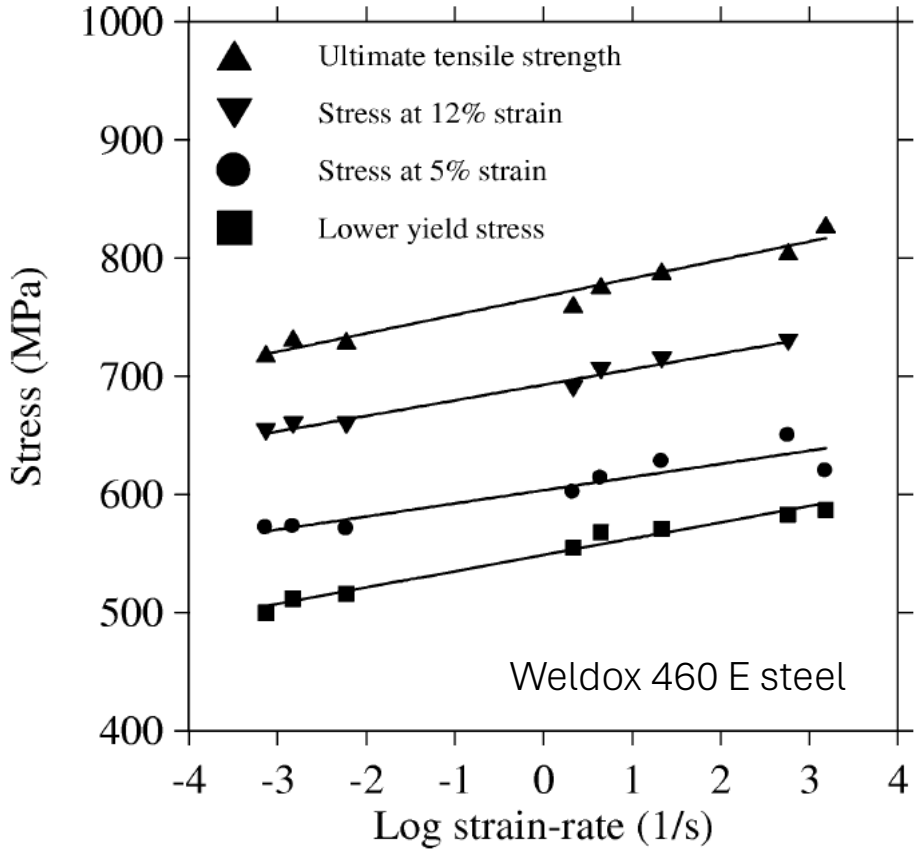
- Consistency condition $(\dot{\lambda}\dot{f} = 0)$ is <u>not valid</u> in viscoplasticity, since during plastic loading we have that $f > 0$ when $\dot{\lambda} > 0$.

Consequence:
$\dot{\lambda}$ must be defined through an equation – a constitutive relation!

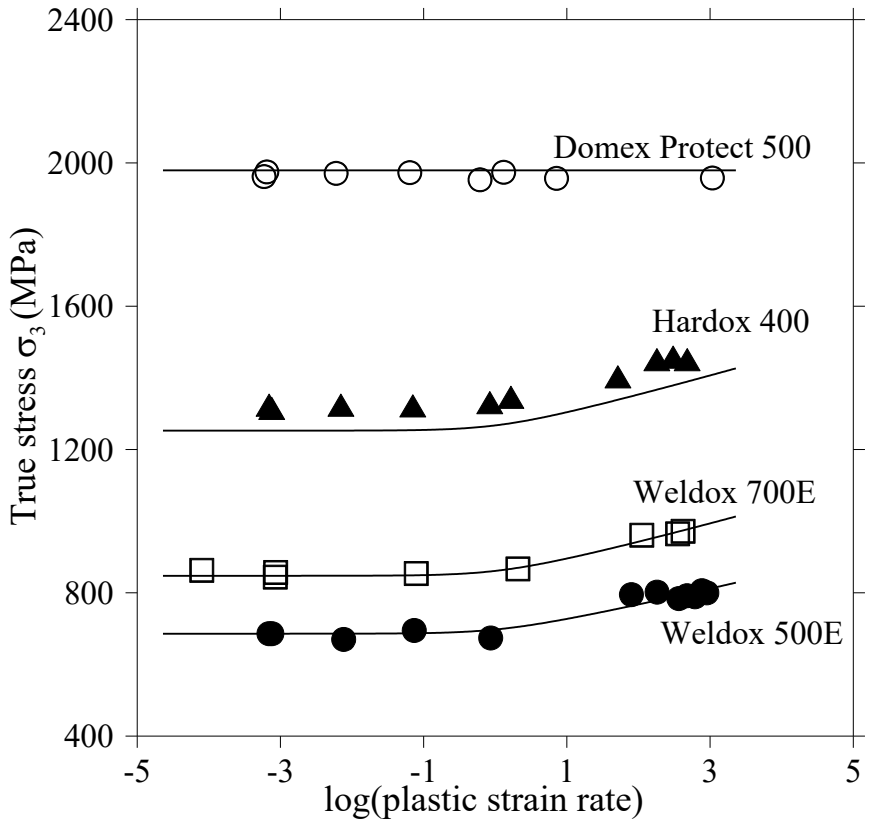




Typical behaviour

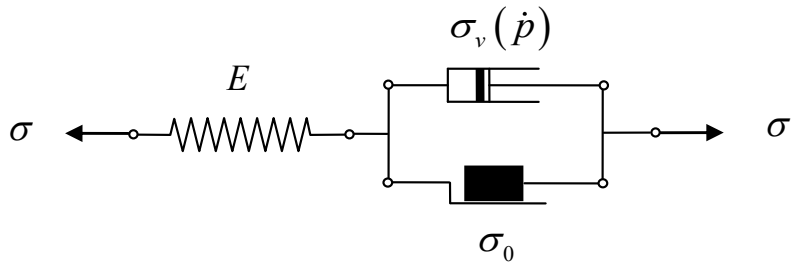


True stress at different plastic strain levels versus strain rate for Weldox 460E steel.

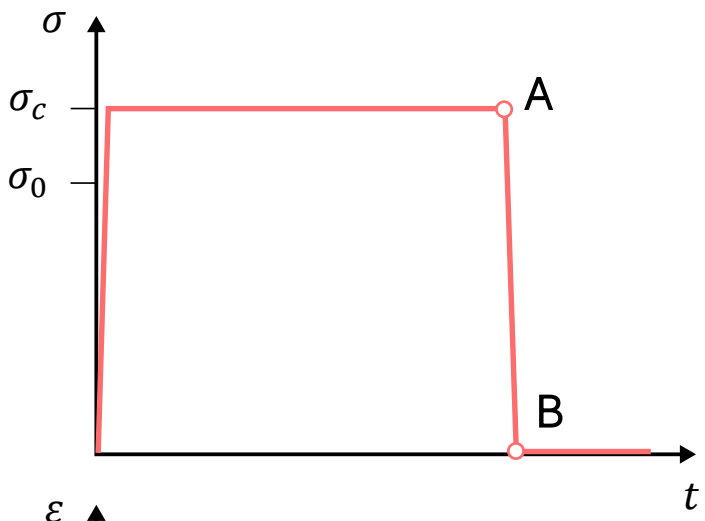


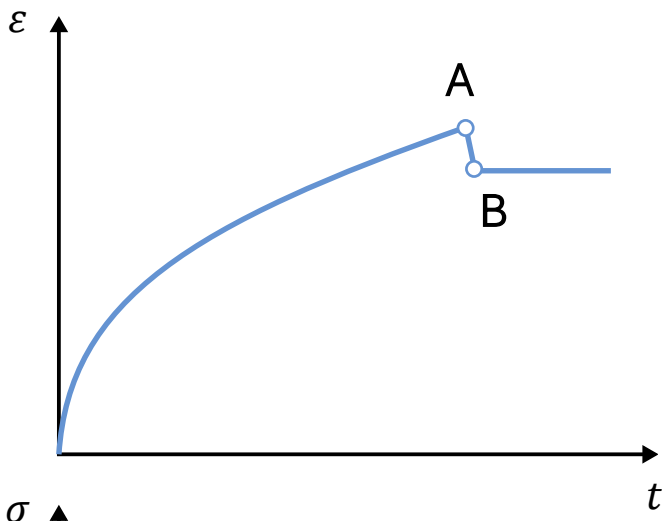
True stress at 3% plastic strain versus strain rate for four steel alloys.

Creep and stress relaxation

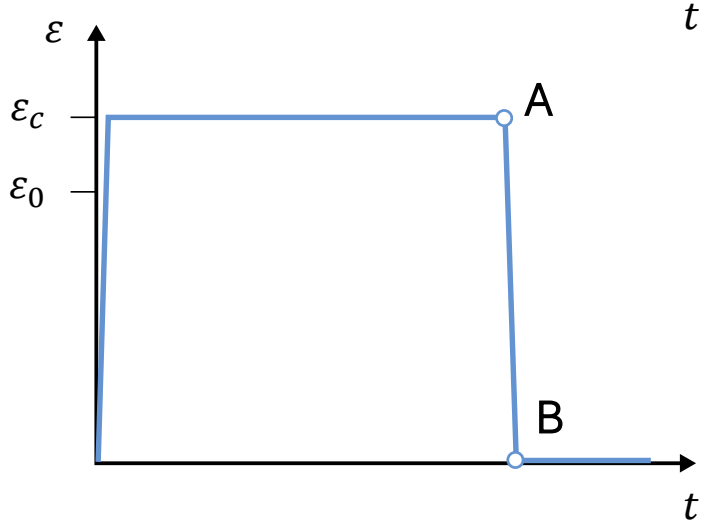


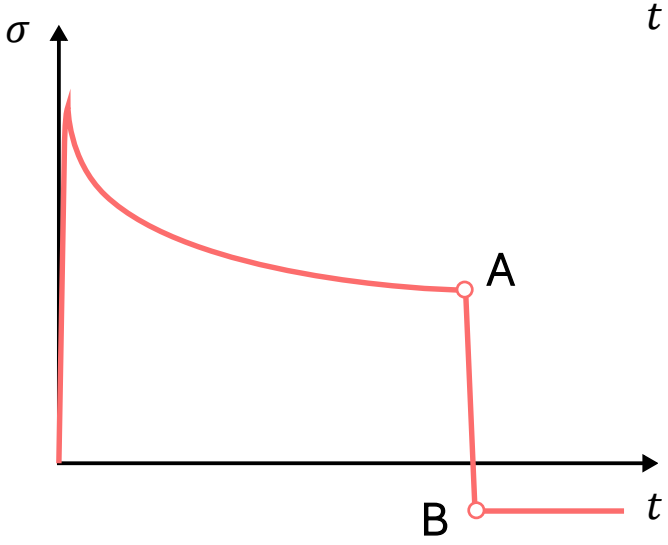
Creep



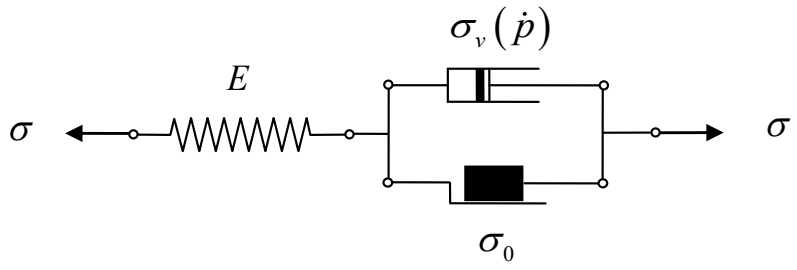


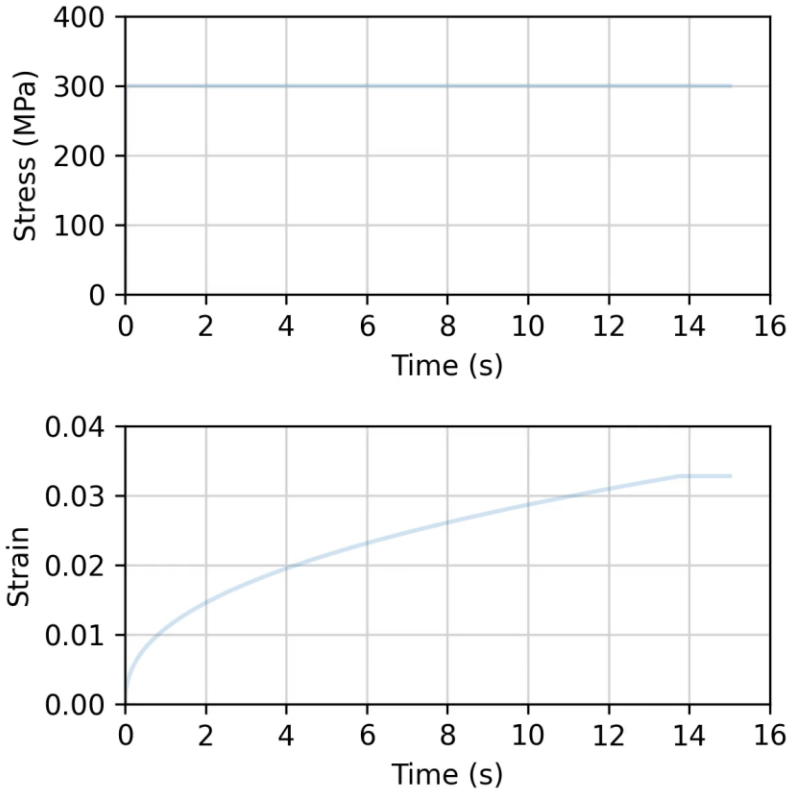
Stress relaxation

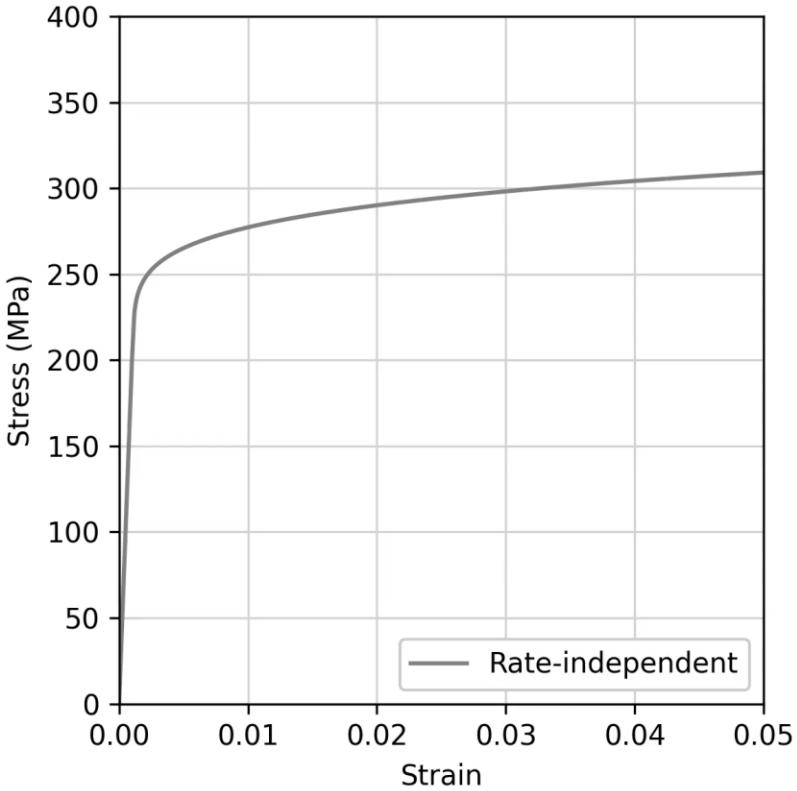


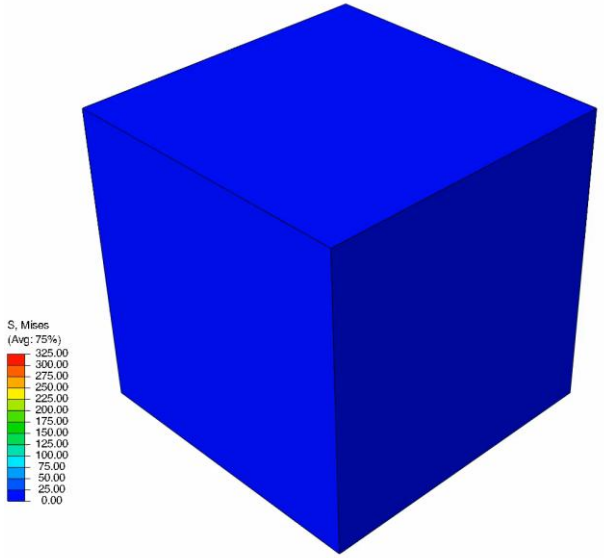


Example: creep





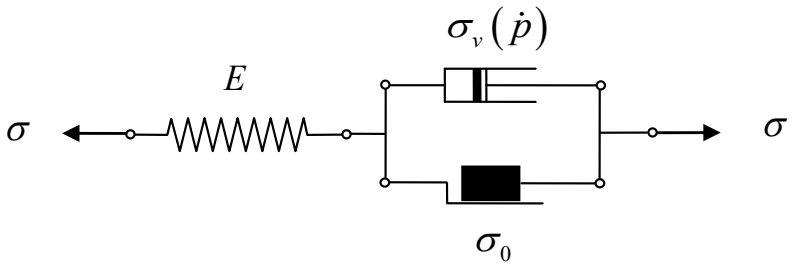


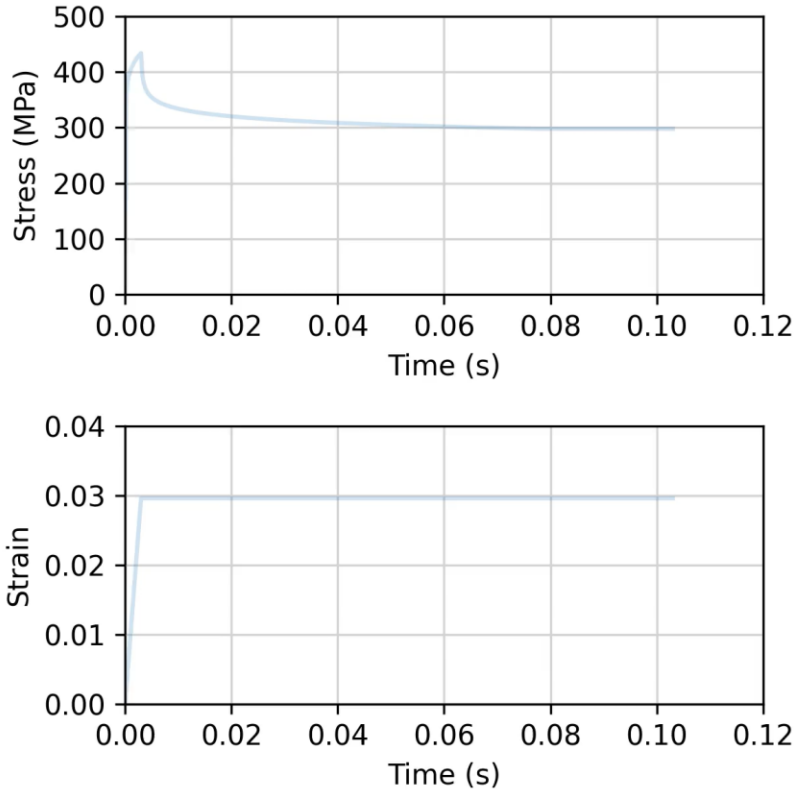


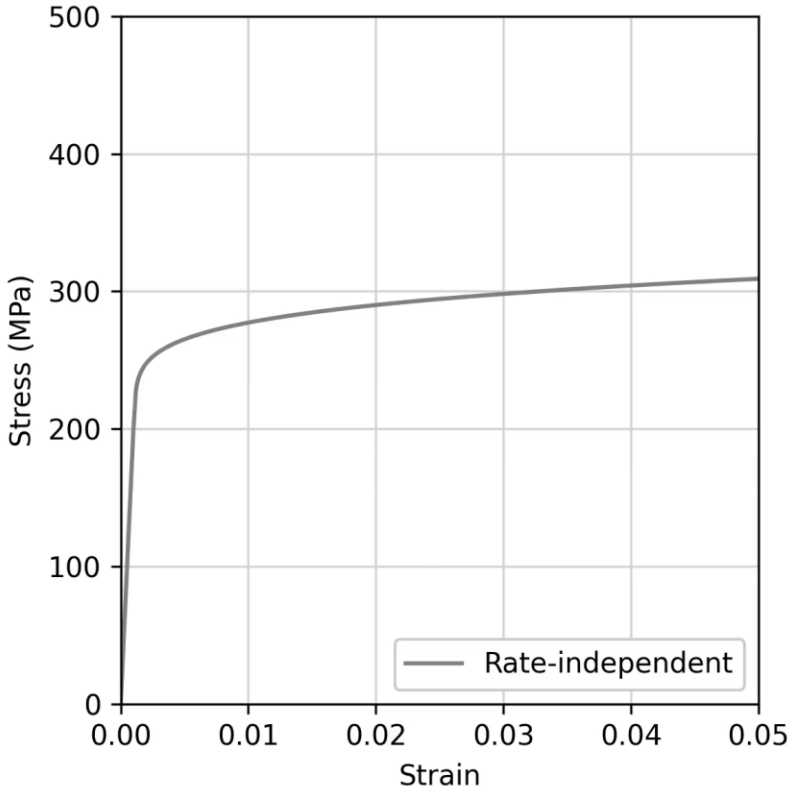
$$\sigma_{\mathrm{eq}} = (A + Bp^n)\left(1 + C\ln\left(1 + \frac{\dot{p}}{\dot{p}_0}\right)\right)$$

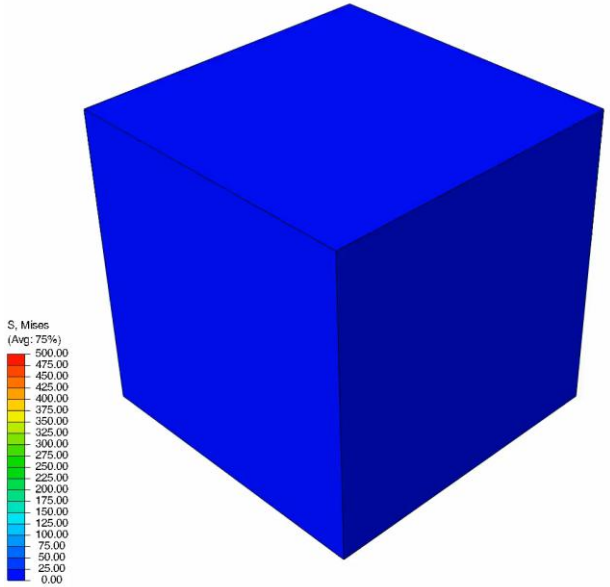
$$A = 200\ \mathrm{MPa}, B = 200\ \mathrm{MPa}, \dot{p}_0 = 0.001\ \mathrm{s}^{-1}, C = 0.05$$

Example: stress relaxation









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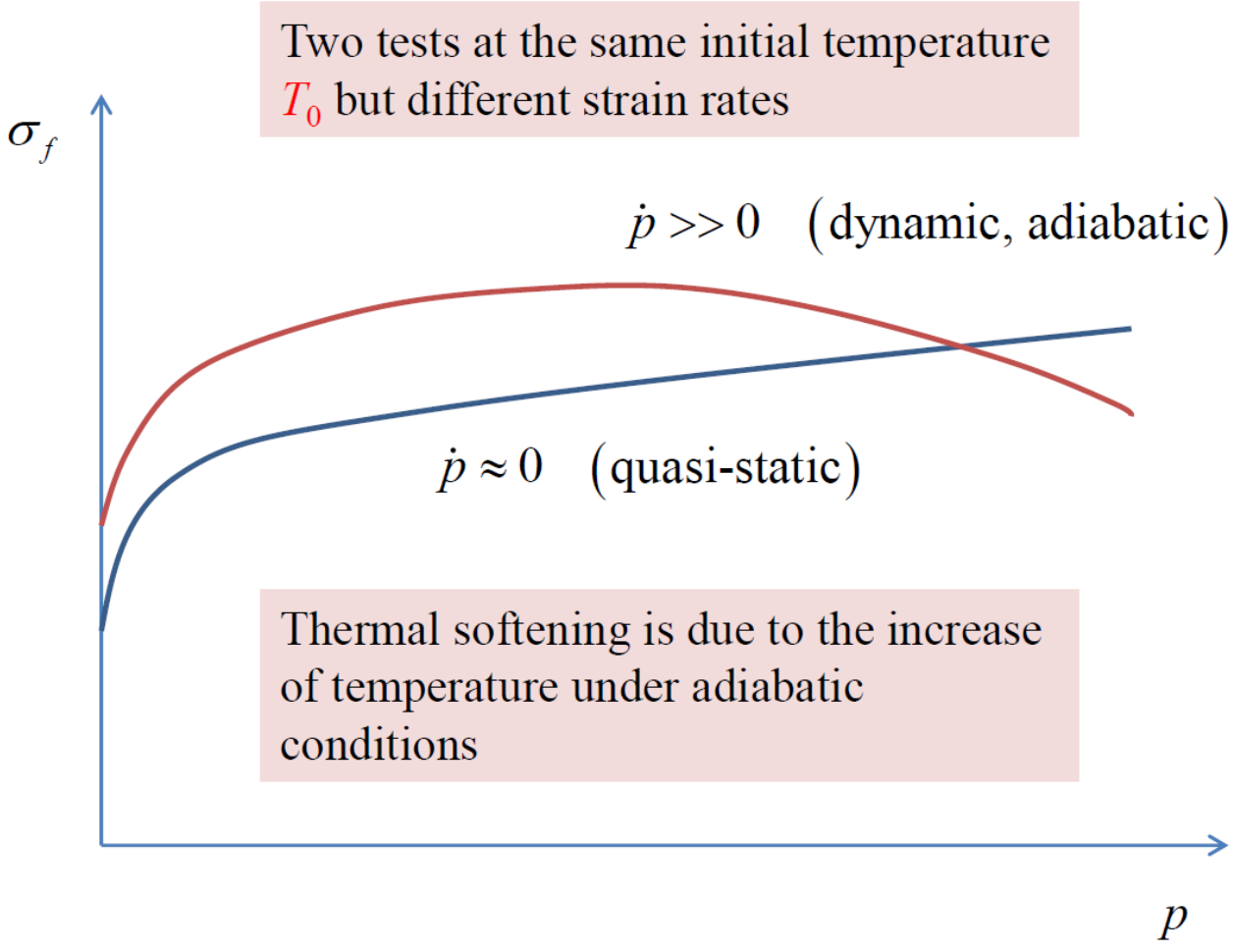
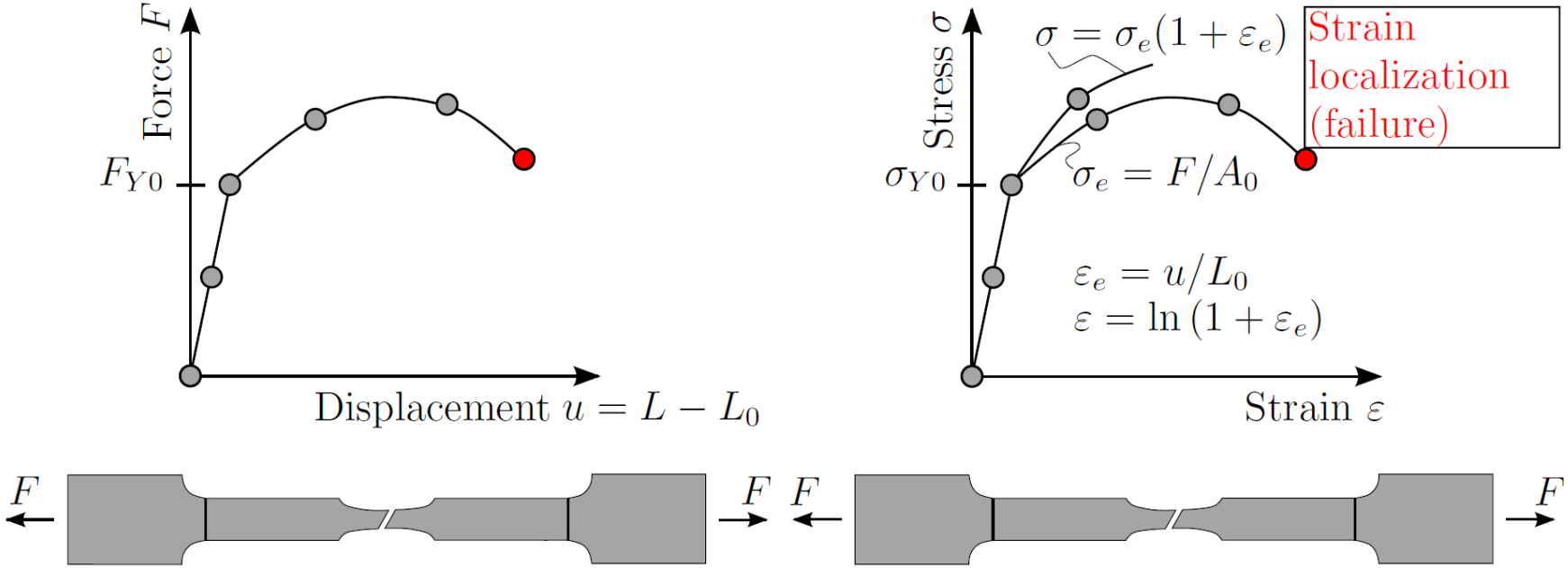
Influence of adiabatic heating



Large deformations

Material behaviour of metals

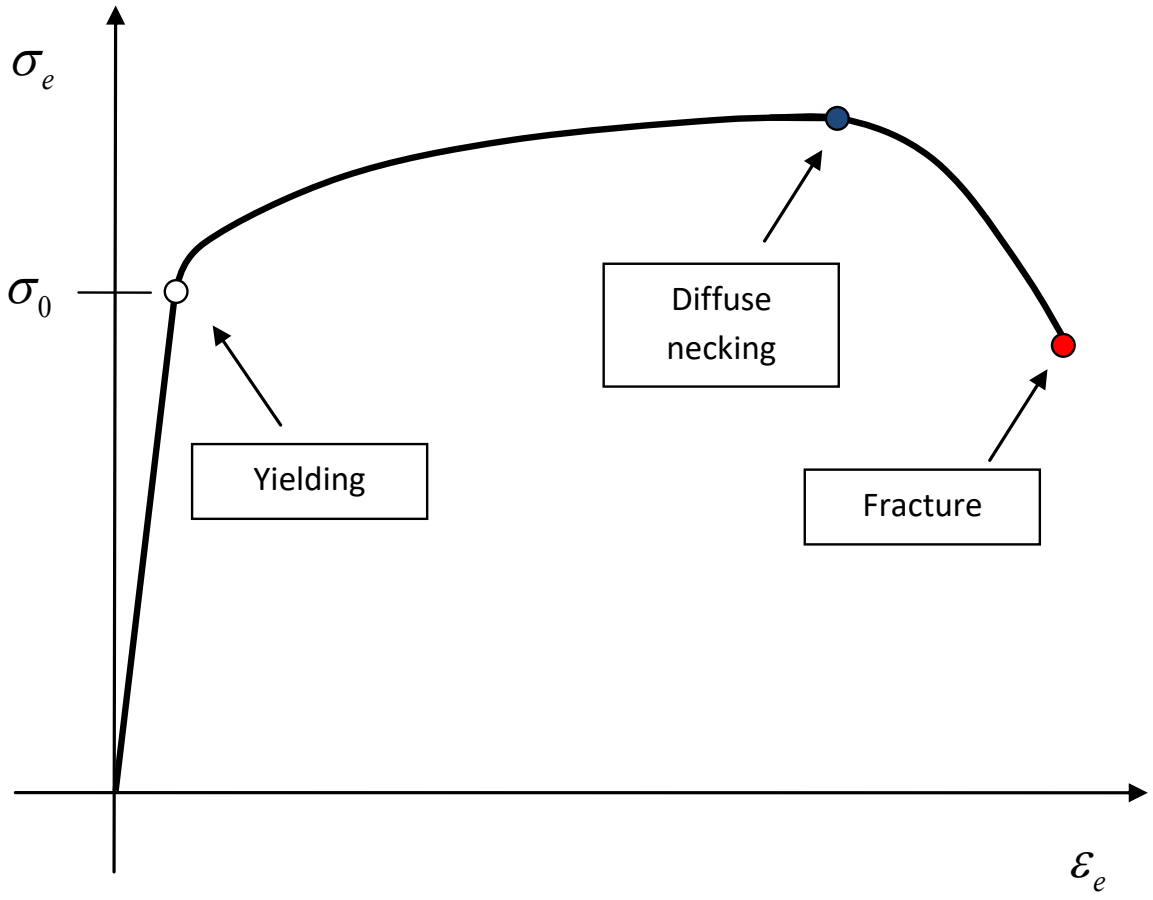
Determined based on uniaxial tensile testing

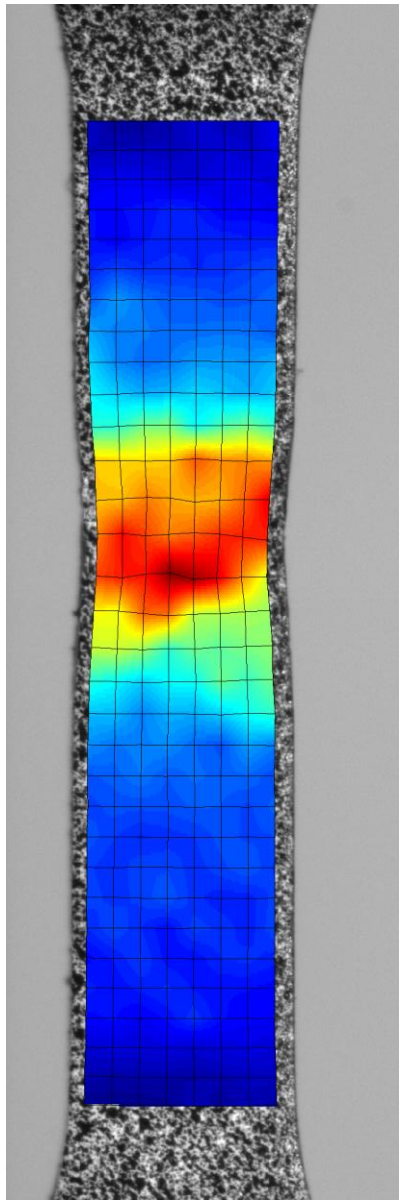


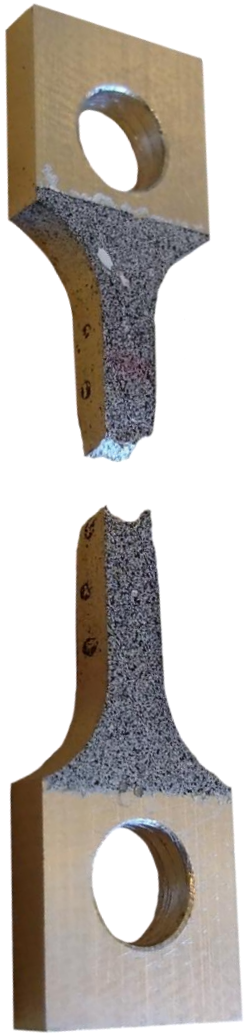
High strength and high elongation before failure

Tensile response

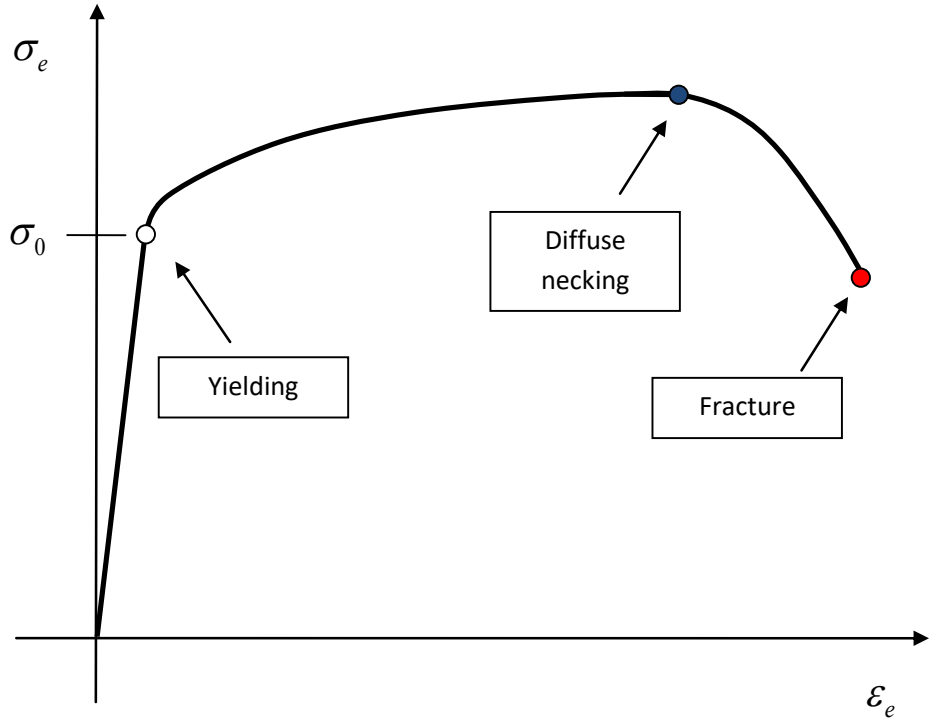
$$\sigma_{\mathrm{e}} = \frac{F}{A_0} = \frac{\sigma_{\mathrm{t}} A}{A_0}$$

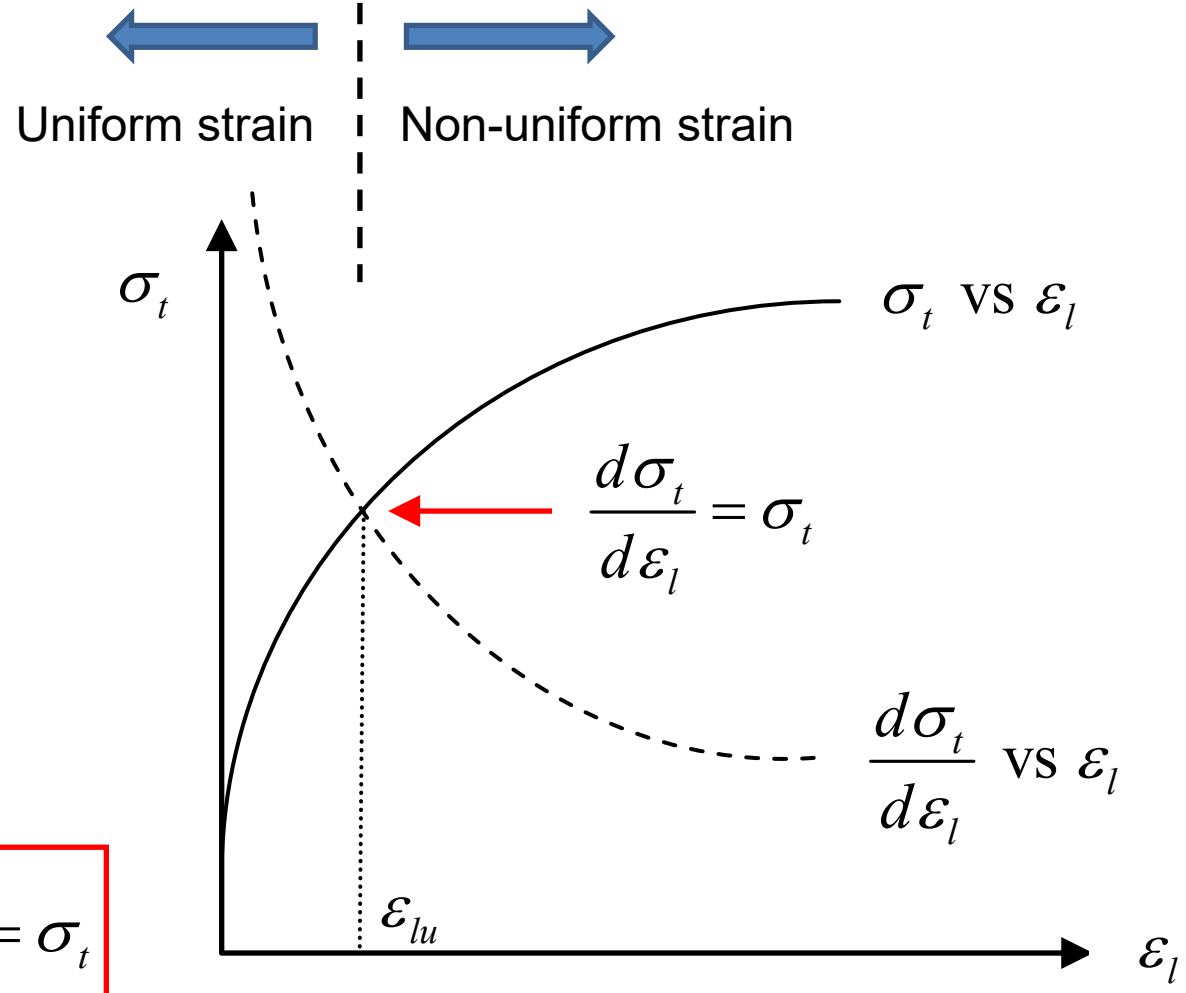






Diffuse necking

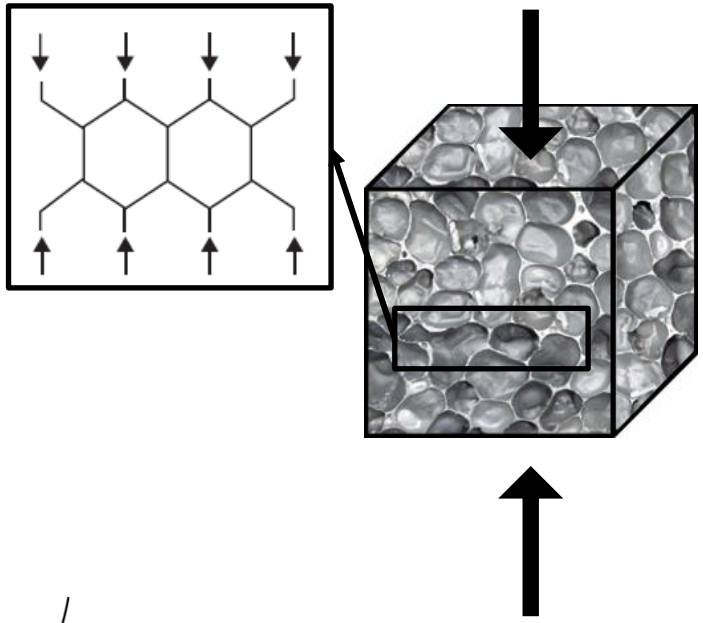




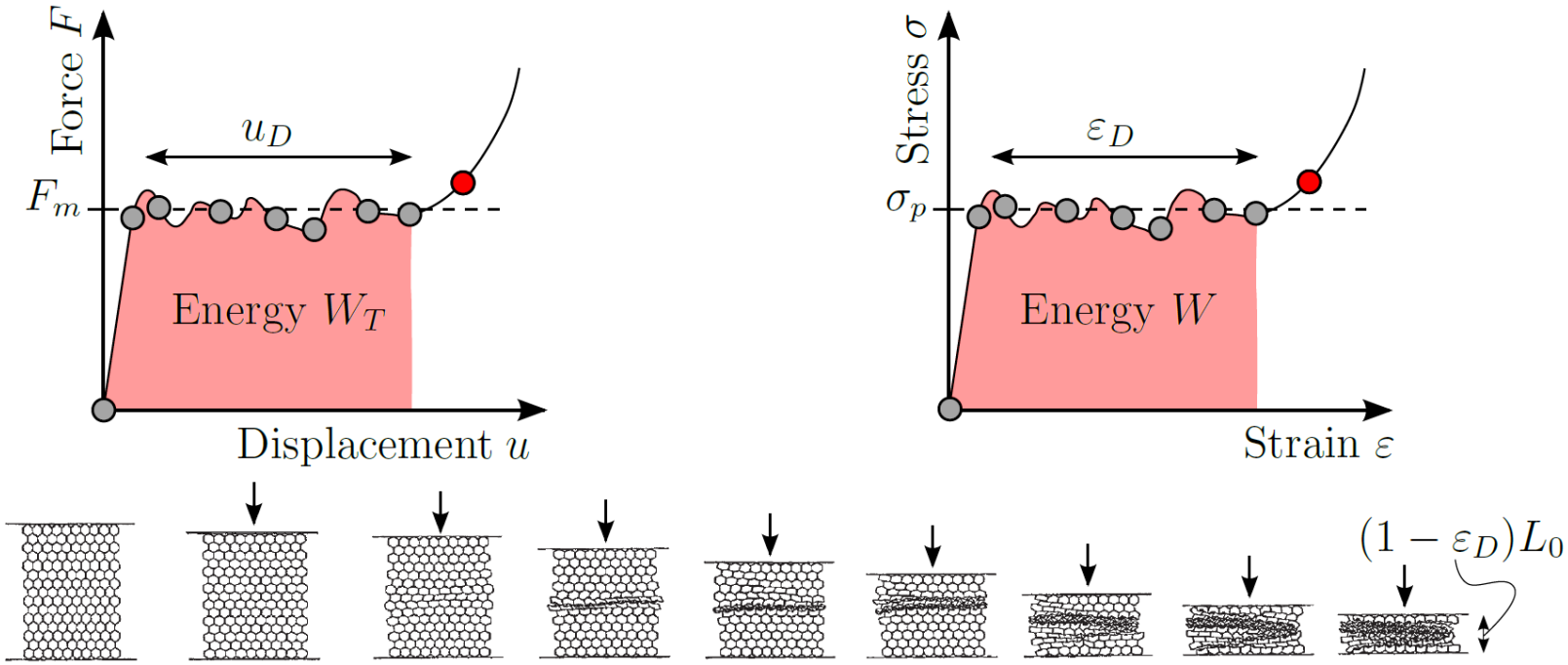
- Necking criterion: $\boxed{d\sigma_e = 0}$ $\Rightarrow$ $\boxed{\frac{d\sigma_t}{d\varepsilon_l} = \sigma_t}$

Metal foams

Behaviour of metal foams



Determined based on uniaxial compression tests

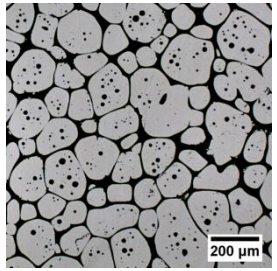


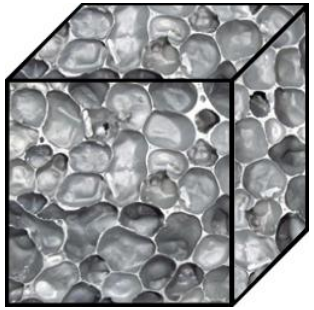
σ_p and ε_D govern the energy absorption until densification (locking)

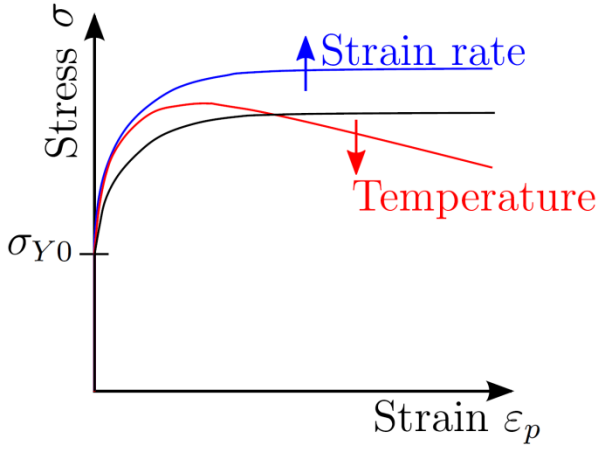
Lu and Yu (2003)

Properties governing material behaviour

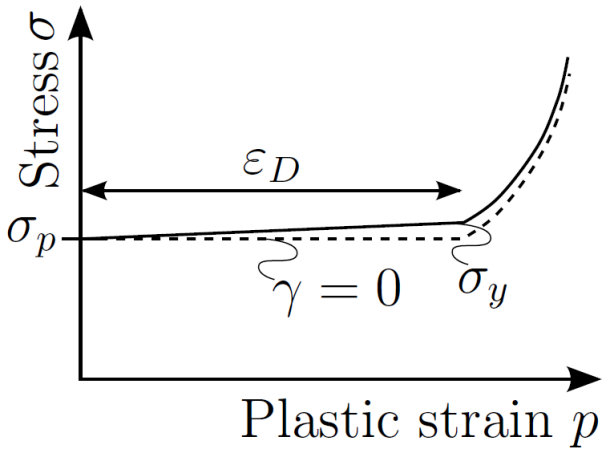
- Metals
  - Microstructure
  - Chemical composition
- Metal foam
  - Material of cell walls
  - Cell topology and shape (microstructure)
  - Relative density ($\varepsilon_D = 1 - \rho_f/\rho_c$)
- Impact and fast transient dynamics
  - Plastic behaviour
  - Rate dependent
  - Temperature softening (large plastic strains)







Modelling of metal foams



- Stresses updated based on the yield function

$$f(\boldsymbol{\sigma}) = \sigma_{eq} - \sigma_y$$

- Isotropic and pressure-sensitive yield criterion

$$\sigma_{eq}(\boldsymbol{\sigma}) = \frac{1}{\sqrt{1+(\alpha/3)^2}}\sqrt{\left(\frac{3}{2}\boldsymbol{\sigma}':\boldsymbol{\sigma}'\right)+\alpha^2\sigma_H^2}$$

- Deshpande-Fleck (stress versus strain)

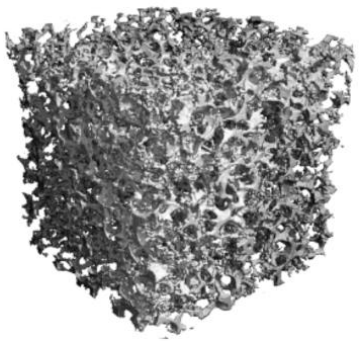
$$\sigma_Y = \sigma_P + \gamma\frac{p}{\varepsilon_D} + \alpha_2 \ln\left(\frac{1}{1-(p/\varepsilon_D)^\beta}\right)$$

Deshpande and Fleck (2000)
Hanssen et al. (2002)
Reyes et al. (2003)

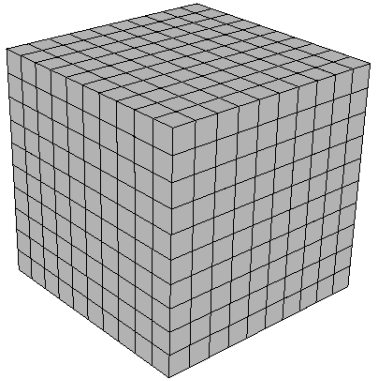
Modelling of metal foams

- Macroscopic level
  - Visible length scales
  - Represent experimental observations of global behaviour
- Microscopic level
  - Length scale not visible
  - Modelling cell topology and material
  - Describe the underlying deformation mechanisms

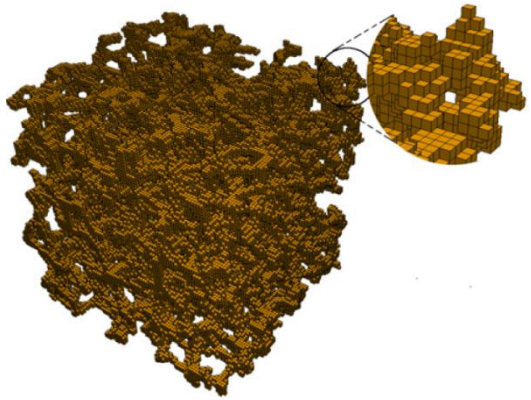
Jung et al. (2015)



Actual sample



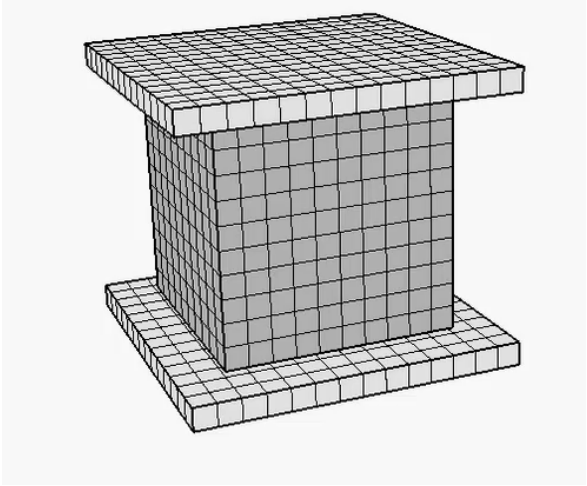
Macroscopic level
(ΔL = 1 cm)



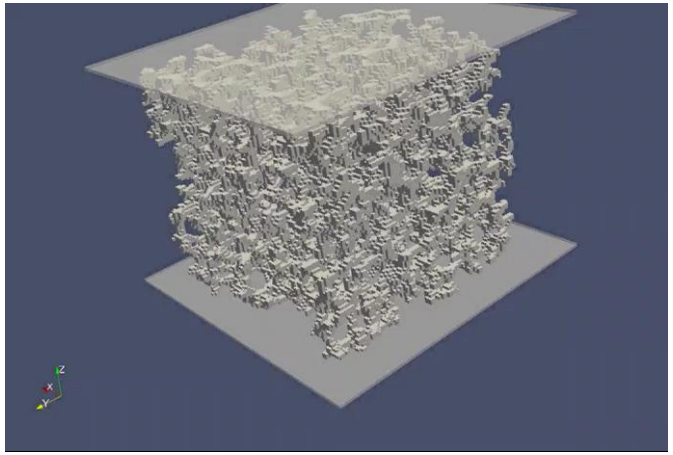
Microscopic level
(ΔL = 170 μm)

Example: Uniaxial compression test

- Macroscopic level
  - Deshpande-Fleck material model
- Microscopic level
  - Modified Johnson-Cook material model



Macroscopic level
(ΔL = 1 cm)



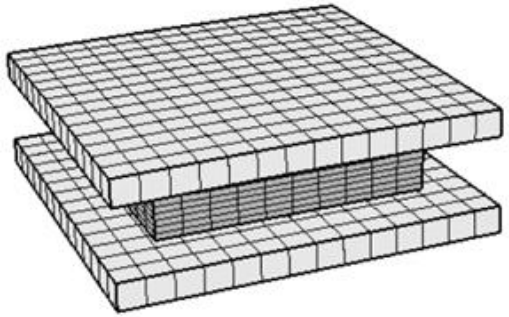
Microscopic level
(ΔL = 170 μm)

Jung et al. (2015)

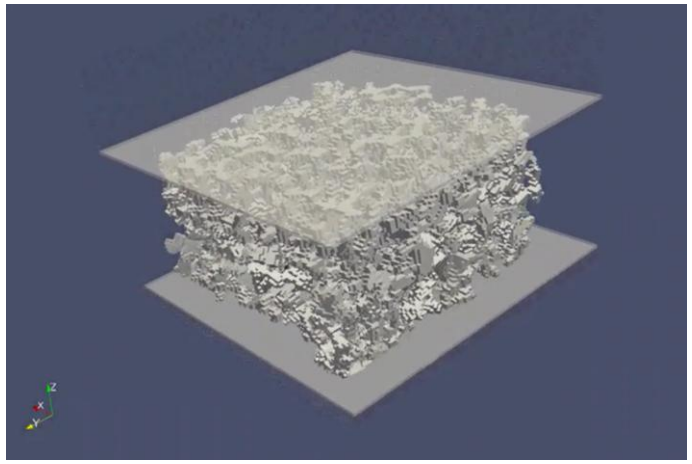
Example: Uniaxial compression test

- Macroscopic level
  - Deshpande-Fleck material model
- Microscopic level
  - Modified Johnson-Cook material model

CPU expensive!



Macroscopic level
(ΔL = 1 cm)



Microscopic level
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Jung et al. (2015)